

Diffusion regime for high-frequency vibrations of randomly heterogeneous structures

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The evolution of the high-frequency vibrational energy density of slender heterogeneous structures such as Timoshenko beams or thick shells is depicted by transport equations or radiative transfer equations (RTEs) in the presence of random heterogeneities. A diffusive regime arises when their correlation lengths are comparable to the wavelength, among other possible situations, and waves are multiply scattered. The purpose of this paper is to expound how diffusion approximations of the RTEs for elastic structures can be derived and to discuss the relevance of the vibrational conductivity analogy invoked in the structural acoustics literature. Its main contribution is the consideration of a heterogeneous background medium with varying parameters and the effects of polarization of elastic waves. The paper also outlines some of the remarkable features of the diffusive regime: depolarization of waves, energy equipartition, and asymptotic Fick's law.

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I. INTRODUCTION

The analysis of high-frequency vibrations of elastic structures has been the subject of intense research for the past decades either in the applied mathematics or the engineering mechanics literature. This issue has a great engineering relevance in the automotive or aerospace industries, for example, for the prediction of the vibroacoustic behavior of complex systems under broadband mechanical or acoustical excitations. The best known and oldest technique is the method of geometrical optics, or WKB ansatz, for which an extensive literature exists: see Refs. 1–3 for some classical references in structural dynamics. This theory shows that the high-frequency energy density of a wave field is the solution of a transport equation. Similar results are obtained by an alternative approach accounting for the cumulative effects of multiple scattering in random media, as done by Stott,⁴ Weaver,⁵ or Margerin,⁶ among others, for electromagnetic, acoustic, or elastic waves. They are also obtained in the frame of the (rigorous) mathematical theory of microlocal analysis developed in the late 1980s and early 1990s by several researchers^{7–10} and extended to random media in Refs. 11–15.

In the limit of small transport mean free paths, transport equations can be approached by diffusion equations.^{16,17} This is the sense of the diffusion approximation originally used by Rybak¹⁸ for elastic waves in a randomly heterogeneous plate. The author derived a diffusion equation for the associated energy density, and this result was used later on by some Russian researchers whose papers were cited by Nefske and Sung in their formulation of the vibrational conductivity analogy (VCA) of structural dynamics.¹⁹ But in this latter publication, it appears that the conditions which have led to the diffusion approximation are no longer stated. Again, it was obtained in a random medium for multiply scattered waves or can be obtained in a *bounded* homogeneous me-

dium for multiply reflected waves: this is the diffuse field approximation used in room acoustics or in the statistical energy analysis (SEA) of engineering structures²⁰ (see also Ref. 21, for example, for an analysis of finite plates with this assumption). For a homogeneous medium, the transport mean free path is infinite and a diffusive regime can theoretically never be reached. Despite its lack of theoretical ground, VCA has been the subject of several investigations in the structural acoustics literature, but the latter were always limited to simple homogeneous structures, typically rods, beams, and more rarely plates; see, for example, Refs. 22–24. More recently, attempts have been made to fit the band-limited smoothed late-time mean square response of large undamped structures to the solution of a diffusion equation of which parameters depend on the early-time response, with a constraint of energy equipartition between modes at late times.²⁵ In the proposed analysis, disorder is taken into account through random generalized forces and observabilities—the projections of the spatial source and receiver filters on the eigenmodes, respectively. The purpose of this paper is to establish, from the (formal) theory of transport of high-frequency waves in randomly heterogeneous media,^{11,12} the correct derivation of diffusion approximations in order to clarify the range of validity, if any, of the VCA and to a lesser extent SEA. The need of such a clarification is also strongly motivated by observations of the diffuse behavior of elastic waves in seismology,¹¹ nondestructive evaluation,²⁶ or structural dynamics.²⁷

The analysis presented here considers random heterogeneities at the same length scale as the wavelength. However, diffusive regimes also arise for random heterogeneities at an intermediate scale between the scale of oscillations and the size of the medium, or for strongly forward-peaked scattering, for example. The more general theory of transport applies to heterogeneous media, does not neglect the possible interferences between energetic propagation modes, and does

not rely on the artificial use of time and/or spatial averages, contrary to what is done in the VCA literature. The derivation of diffusion approximations for linear or nonlinear Boltzmann equations has been considered for several decades and the related literature is too vast to be listed exhaustively here. Some popular approaches are the Hilbert expansion method, e.g., Refs. 28–30, or stochastic averaging, e.g., Ref. 31. In Sec. II, some basic results pertaining to high-frequency wave propagation in a heterogeneous elastic slender structure are summarized. The linear Boltzmann equations or radiative transfer equations (RTEs) which characterize its energy density evolution are introduced. The presentation focuses on the case of thick shells and then beams, but all results apply to plates as well. RTEs for slender structures form a system of multigroup transport equations which is discussed in Sec. III. The collision operators on its right-hand side are projected on the unit sphere with respect to the direction variable. This step identifies the collision process, which mixes wave directions until the memory of initial wave fronts is lost. Such a situation, together with wave depolarization, characterizes a diffusive behavior. The main contribution of the paper is Sec. IV, where the derivation of diffusion equations from multigroup transport equations in the limit of small scattering mean free paths is outlined. The analysis is based on a formal Hilbert expansion, which can be made rigorous as in Refs. 28–30, for example, and uses some crucial algebraic properties of the collision operators. It extends the previous conclusions of Refs. 32–34 for a homogeneous elastic medium to a heterogeneous background medium with varying wave speeds. Apart from losing the memory of their initial directions of propagation, transported waves in the diffusive regime also become approximately depolarized and their energy densities equally partitioned; the proposed analysis clearly exhibits this property. These results are summarized and briefly commented in Sec. V.

II. HIGH-FREQUENCY ENERGY EVOLUTION IN SLENDER STRUCTURES

This section summarizes the high-frequency energy evolution properties of slender structures as derived in Ref. 33 on the basis of the kinetic models developed in Refs. 10–12. The setting presented here for polarized elastic waves is in many respects similar to the setting of Ref. 35 for electromagnetic waves.

A. Shells

Let us consider a thick curved shell occupying the domain $\Sigma \times (-h/2, h/2)$ of $\mathbb{R}^3$, where the (regular) mean surface Σ is parametrized by its curvilinear coordinates $\mathbf{x} = (x, y) \in \mathbb{R}^2$ and $h > 0$ is its thickness. The associated natural tangent basis is denoted by $(\mathbf{e}_1, \mathbf{e}_2)$ such that $\mathbf{e}_1 = \partial_x \mathbf{x}$ and $\mathbf{e}_2 = \partial_y \mathbf{x}$. Tangent motions of the mean surface are denoted by $\mathbf{u}^\epsilon$, w^ϵ is the radial displacement (deflection), and $\boldsymbol{\theta}^\epsilon$ is the vector of changes of slope of the unit normal $\hat{\mathbf{n}} = \mathbf{e}_1 \times \mathbf{e}_2 / |\mathbf{e}_1 \times \mathbf{e}_2|$ to Σ (see Fig. 1) in the local tangent basis. The overall displacement field of the shell in the local basis $(\mathbf{e}_1, \mathbf{e}_2, \hat{\mathbf{n}})$ thus writes

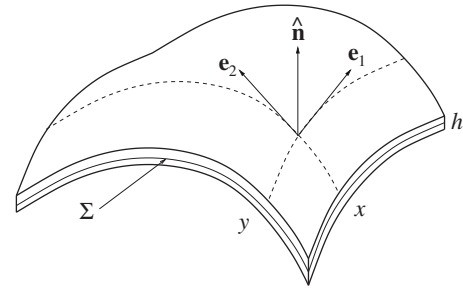


FIG. 1. An elastic shell structure occupying the domain $\Sigma \times (-h/2, h/2)$ of $\mathbb{R}^3$. Σ is the mean surface of the shell, h is its thickness, and $(\mathbf{e}_1, \mathbf{e}_2)$ is its natural tangent basis at the point $\mathbf{x} = (x, y)$. $\hat{\mathbf{n}}$ is the unit normal outward to the mean surface.

$$\mathbf{U}^\epsilon(\mathbf{x}) = \mathbf{u}^\epsilon(\mathbf{x}) - z\boldsymbol{\theta}^\epsilon(\mathbf{x}) + w^\epsilon(\mathbf{x})\hat{\mathbf{n}}(\mathbf{x}), \quad (1)$$

where $-h/2 < z < h/2$ is the radial coordinate along $\hat{\mathbf{n}}$. The above expression corresponds to the first-order approximation of the displacement field of an elastic three-dimensional medium with respect to its thickness, and is nothing but the kinematic assumption which defines a shell. Its second-order normal curvature tensor $\mathbf{B}$ is

$$\mathbf{B} = \sum_{\alpha, \beta=1}^2 B_{\alpha\beta} \mathbf{e}_\alpha \otimes \mathbf{e}_\beta, \quad B_{\alpha\beta} = -\mathbf{e}_\alpha \cdot \partial_\beta \hat{\mathbf{n}} = \hat{\mathbf{n}} \cdot \partial_\beta \mathbf{e}_\alpha$$

and it is symmetric. The mean radius of the shell $R(\mathbf{x}) = (\text{Tr } \mathbf{B})^{-1}$ at each point $\mathbf{x}$ of Σ , where $\text{Tr } \mathbf{B}$ is the sum of the principal curvatures, is assumed to be much larger than the thickness h . The latter is in turn significantly lower than the typical wavelength ϵ of the excitations, which means that functions $\mathbf{u}^\epsilon$, w^ϵ and $\boldsymbol{\theta}^\epsilon$ oscillate at frequencies of order ϵ^{-1} (see below).

The elastic material constituting this shell is characterized by its density $\rho^\epsilon(\mathbf{x})$ and its elasticity tensor $\mathbf{C}^\epsilon(\mathbf{x})$, depending on two parameters if it is isotropic. Let $E^\epsilon(\mathbf{x})$ be its Young's modulus and $\nu^\epsilon(\mathbf{x})$ be its Poisson's coefficient, then Lamé's parameters are $\mu^\epsilon = E^\epsilon/2(1 + \nu^\epsilon)$ (the shear modulus) and $\lambda^\epsilon = 2\nu^\epsilon\mu^\epsilon/1 - 2\nu^\epsilon$, and the constitutive equation is written as

$$\mathbf{C}^\epsilon : \boldsymbol{\epsilon} = \lambda^\epsilon (\text{Tr } \boldsymbol{\epsilon}) \mathbf{I} + 2\mu^\epsilon \boldsymbol{\epsilon}$$

for the second order linearized strain tensor $\boldsymbol{\epsilon}$. Using, for example, the model of Naghdi and Cooper,³⁶ the membrane forces tensor $\mathbf{N}^\epsilon$, shear forces $\mathbf{T}^\epsilon$, and bending moments tensor $\mathbf{M}^\epsilon$ are given by

$$\mathbf{N}^\epsilon = h\mathbf{C}^\epsilon : (\boldsymbol{\epsilon}(\mathbf{u}^\epsilon) - w^\epsilon \mathbf{B}), \quad \mathbf{T}^\epsilon = \mu_r^\epsilon h (\nabla_\Sigma w^\epsilon - \boldsymbol{\theta}^\epsilon - \mathbf{B}\mathbf{u}^\epsilon), \quad (2)$$

$$\mathbf{M}^\epsilon = \frac{h^3}{12} \mathbf{C}^\epsilon : \boldsymbol{\epsilon}(\boldsymbol{\theta}^\epsilon),$$

where $\boldsymbol{\epsilon}(\mathbf{u}) = \nabla_\Sigma \otimes_s \mathbf{u}$, $\otimes_s$ is the symmetrized tensor product, and $\nabla_\Sigma = (\partial_x, \partial_y)$ is the gradient on the mean surface. $\mu_r^\epsilon = \kappa^\epsilon \mu^\epsilon$ is the reduced shear modulus, where $\kappa^\epsilon < 1$ is the reduced shear coefficient which depends on Poisson's ratio. The equations of motion are

$$\begin{aligned}\rho^\epsilon \partial_t^2 \mathbf{u}^\epsilon &= \text{div}_\Sigma \mathbf{N}^\epsilon + \mathbf{B} \mathbf{T}^\epsilon, & \rho^\epsilon \partial_t^2 w^\epsilon &= \text{div}_\Sigma \mathbf{T}^\epsilon + \mathbf{B} : \mathbf{N}^\epsilon, \\ \frac{\rho^\epsilon h^2}{12} \partial_t^2 \boldsymbol{\theta}^\epsilon &= \text{div}_\Sigma \mathbf{M}^\epsilon + \mathbf{T}^\epsilon.\end{aligned}\quad (3)$$

These evolution equations are completed by initial conditions $(\mathbf{u}^\epsilon, w^\epsilon, \boldsymbol{\theta}^\epsilon, \partial_t \mathbf{u}^\epsilon, \partial_t w^\epsilon, \partial_t \boldsymbol{\theta}^\epsilon)(\mathbf{x}, 0) = (\mathbf{u}_0^\epsilon, w_0^\epsilon, \boldsymbol{\theta}_0^\epsilon, \mathbf{u}_1^\epsilon, w_1^\epsilon, \boldsymbol{\theta}_1^\epsilon)(\mathbf{x})$ which oscillate at frequencies of order ϵ^{-1} , and conservation of the elastic energy if materials are nondissipative writes

$$\begin{aligned}\mathcal{E}^\epsilon(t) &= \frac{1}{2} \int_\Sigma [\mathbf{N}^\epsilon : (\boldsymbol{\epsilon}(\mathbf{u}^\epsilon) - w^\epsilon \mathbf{B}) + \mathbf{M}^\epsilon : \boldsymbol{\epsilon}(\boldsymbol{\theta}^\epsilon) \\ &\quad + \mathbf{T}^\epsilon \cdot (\nabla_\Sigma w^\epsilon - \boldsymbol{\theta}^\epsilon - \mathbf{B} \mathbf{u}^\epsilon)] d\mathbf{x} = \mathcal{E}^\epsilon(0).\end{aligned}\quad (4)$$

Kinetic models such as transport or radiative transfer are aimed at describing the evolution of the elastic energy *density*:

$$\begin{aligned}\mathcal{E}^\epsilon(\mathbf{x}, t) &= \frac{1}{2} [\mathbf{N}^\epsilon : (\boldsymbol{\epsilon}(\mathbf{u}^\epsilon) - w^\epsilon \mathbf{B}) + \mathbf{M}^\epsilon : \boldsymbol{\epsilon}(\boldsymbol{\theta}^\epsilon) \\ &\quad + \mathbf{T}^\epsilon \cdot (\nabla_\Sigma w^\epsilon - \boldsymbol{\theta}^\epsilon - \mathbf{B} \mathbf{u}^\epsilon)]\end{aligned}\quad (5)$$

and power flow *density*:

$$\Pi^\epsilon(\mathbf{x}, t) = -(\mathbf{N}^\epsilon \partial_t \mathbf{u}^\epsilon + \mathbf{M}^\epsilon \partial_t \boldsymbol{\theta}^\epsilon + \mathbf{T}^\epsilon \partial_t w^\epsilon) \quad (6)$$

in the high-frequency limit $\epsilon \rightarrow 0$, when material parameters $\rho^\epsilon(\mathbf{x})$, $\lambda^\epsilon(\mathbf{x})$, and $\mu^\epsilon(\mathbf{x})$ are randomly varying at a length scale comparable with the typical wavelength ϵ . More precisely, it is assumed that

$$\begin{aligned}\rho^\epsilon(\mathbf{x}) &= \underline{\rho}(\mathbf{x}) \left[1 + \sqrt{\epsilon} X_1 \left(\frac{\mathbf{x}}{\epsilon} \right) \right], \\ \frac{1}{\lambda^\epsilon(\mathbf{x})} &= \frac{1}{\underline{\lambda}(\mathbf{x})} \left[1 + \sqrt{\epsilon} X_2 \left(\frac{\mathbf{x}}{\epsilon} \right) \right], \\ \frac{1}{\mu^\epsilon(\mathbf{x})} &= \frac{1}{\underline{\mu}(\mathbf{x})} \left[1 + \sqrt{\epsilon} X_3 \left(\frac{\mathbf{x}}{\epsilon} \right) \right], \\ \frac{1}{\mu_r^\epsilon(\mathbf{x})} &= \frac{1}{\underline{\mu}_r(\mathbf{x})} \left[1 + \sqrt{\epsilon} X_4 \left(\frac{\mathbf{x}}{\epsilon} \right) \right],\end{aligned}\quad (7)$$

where $(X_j(\mathbf{x}), \mathbf{x} \in \mathbb{R}^2)$, $j=1, \dots, 4$, are four real-valued statistically homogeneous and centered second-ordered stochastic processes with correlation lengths comparable to the wavelength in the medium. $\underline{\rho}(\mathbf{x}) = \mathbb{E}\{\rho^\epsilon(\mathbf{x})\}$, $\underline{\lambda}(\mathbf{x}) = \mathbb{E}\{\lambda^\epsilon(\mathbf{x})^{-1}\}^{-1}$, $\underline{\mu}(\mathbf{x}) = \mathbb{E}\{\mu^\epsilon(\mathbf{x})^{-1}\}^{-1}$, and $\underline{\mu}_r(\mathbf{x}) = \mathbb{E}\{\mu_r^\epsilon(\mathbf{x})^{-1}\}^{-1}$ are averaged macroscopic density and Lamé's moduli, respectively, $\mathbb{E}\{X\}$ standing for the mathematical expectation of a random variable X . The perturbation amplitudes in the above model are chosen as the square root of the normalized small wavelength ϵ in order to avoid localization if they were too large, while this scaling still allows the energy within the medium to be significantly modified by material fluctuations.^{11,12,14} In this setting, the effect of scattering on heterogeneities is comparable to the effect of the slowly varying background medium, evading localization at this scale. Equation (7) is general enough to embed the occurrence of discrete scatterers, in

which case random processes $(X_j(\mathbf{x}), \mathbf{x} \in \mathbb{R}^2)$ are simply random characteristic functions, as in, e.g., Ref. 37. Note that Lamé's and reduced shear moduli depend nonlinearly on Young's modulus and Poisson's ratio, thus random variables X_2 , X_3 , and X_4 are *a priori* strongly correlated but the above model does not enforce any quantification of these correlations.

Then it is shown in Ref. 33, based on the works of Refs. 11 and 12, that there exist a positive scalar w_T and Hermitian, positive two-by-two matrices $\mathbf{W}_P$ and $\mathbf{W}_S$ such that

$$\begin{aligned}\lim_{\epsilon \rightarrow 0} \mathcal{E}^\epsilon(\mathbf{x}, t) &= \int_{\mathbb{R}^2} w_T(\mathbf{x}, \mathbf{k}, t) d\mathbf{k} \\ &\quad + \sum_{\alpha=P,S} \int_{\mathbb{R}^2} \text{Tr} \mathbf{W}_\alpha(\mathbf{x}, \mathbf{k}, t) d\mathbf{k},\end{aligned}\quad (8)$$

whence w_T , $\text{Tr} \mathbf{W}_P$, and $\text{Tr} \mathbf{W}_S$ correspond to phase-space energy densities in $T^*\mathbb{R}^2 \equiv \mathbb{R}_x^2 \times \mathbb{R}_k^2$. They are associated with pure transverse waves propagating at an average speed $c_T = \sqrt{\mu_r h / \rho}$ and polarized normally to the mean surface Σ , and to compressional and shear waves propagating at the average speeds $c_P = \sqrt{(\lambda + 2\mu)h / \rho}$ and $c_S = \sqrt{\mu h / \rho}$, respectively, and polarized in Σ . Equation (8) states that these energy densities must be added up for all possible directions of propagation $\hat{\mathbf{k}} := \mathbf{k} / |\mathbf{k}|$ and all wavenumbers $|\mathbf{k}| > 0$. Compressional waves in shells are generated by either in-plane motions or pure bending motions polarized along the propagation direction $\hat{\mathbf{k}}$, as seen from the kinematic definition (1): $\mathbf{u}^\epsilon \cdot \hat{\mathbf{k}}$ is the in-plane contribution, and $-z \boldsymbol{\theta}^\epsilon \cdot \hat{\mathbf{k}}$ is the pure bending contribution. This induces a two-by-two matrix structure of the compressional energy coherence matrix $\mathbf{W}_P$. Its diagonal entries are the energy densities themselves and its off-diagonal entries characterize all interactions between both types of motion just like the Stokes parameters used for the description of light polarization. Likewise, shear waves are generated by in-plane and pure bending motions polarized perpendicularly to the propagation direction $\hat{\mathbf{k}}$, hence the two-by-two matrix structure of the shear energy coherence matrix $\mathbf{W}_S$ as well. The interpretation of its diagonal and off-diagonal entries is the same as for $\mathbf{W}_P$. Furthermore, w_T , $\mathbf{W}_P$, and $\mathbf{W}_S$ satisfy a system of coupled RTEs written as

$$\begin{aligned}\partial_t w_T + \{\lambda_T, w_T\} &= \int_{\mathbb{R}^2} \sigma_{TT}(\mathbf{x}, \mathbf{k}, \mathbf{k}') w_T(\mathbf{x}, \mathbf{k}', t) \delta(\lambda_T(\mathbf{x}, \mathbf{k}) \\ &\quad - \lambda_T(\mathbf{x}, \mathbf{k}')) d\mathbf{k}' - \Sigma_T(\mathbf{x}, \mathbf{k}) w_T(\mathbf{x}, \mathbf{k}, t),\end{aligned}\quad (9)$$

and

$$\begin{aligned}\partial_t \mathbf{W}_P + \{\lambda_P, \mathbf{W}_P\} &= \int_{\mathbb{R}^2} \sigma_{PP}(\mathbf{x}, \mathbf{k}, \mathbf{k}') \mathbf{W}_P(\mathbf{x}, \mathbf{k}', t) \delta(\lambda_P(\mathbf{x}, \mathbf{k}) - \lambda_P(\mathbf{x}, \mathbf{k}')) d\mathbf{k}' \\ &\quad + \int_{\mathbb{R}^2} \sigma_{PS}(\mathbf{x}, \mathbf{k}, \mathbf{k}') \mathbf{W}_S(\mathbf{x}, \mathbf{k}', t) \delta(\lambda_P(\mathbf{x}, \mathbf{k}) - \lambda_S(\mathbf{x}, \mathbf{k}')) d\mathbf{k}' \\ &\quad - \Sigma_P(\mathbf{x}, \mathbf{k}) \mathbf{W}_P(\mathbf{x}, \mathbf{k}, t),\end{aligned}\quad (10)$$

$$\begin{aligned}
& \partial_t \mathbf{W}_S + \{\lambda_S, \mathbf{W}_S\} \\
&= \int_{\mathbb{R}^2} \sigma_{SS}(\mathbf{x}, \mathbf{k}, \mathbf{k}') \mathbf{W}_S(\mathbf{x}, \mathbf{k}', t) \delta(\lambda_S(\mathbf{x}, \mathbf{k}) - \lambda_S(\mathbf{x}, \mathbf{k}')) d\mathbf{k}' \\
&+ \int_{\mathbb{R}^2} \sigma_{SP}(\mathbf{x}, \mathbf{k}, \mathbf{k}') \mathbf{W}_P(\mathbf{x}, \mathbf{k}', t) \delta(\lambda_S(\mathbf{x}, \mathbf{k}) - \lambda_P(\mathbf{x}, \mathbf{k}')) d\mathbf{k}' \\
&- \Sigma_S(\mathbf{x}, \mathbf{k}) \mathbf{W}_S(\mathbf{x}, \mathbf{k}, t), \tag{11}
\end{aligned}$$

where

$$\lambda_\alpha(\mathbf{x}, \mathbf{k}) = c_\alpha(\mathbf{x})|\mathbf{k}|, \quad \alpha = T, P, S, \tag{12}$$

$\{f, g\} = \nabla_{\mathbf{k}} f \cdot \nabla_{\mathbf{x}} g - \nabla_{\mathbf{x}} f \cdot \nabla_{\mathbf{k}} g$ stands for the usual Poisson bracket, and $\mathbf{u}_1 \cdot \mathbf{u}_2$ stands for the Euclidean scalar product of two vectors $\mathbf{u}_1$ and $\mathbf{u}_2$. The so-called differential scattering cross sections $\sigma_{\alpha\beta}$, $\alpha, \beta = T, P, S$, are positive functions which are known explicitly as functions of the power spectral density functions $\mathbf{k} \mapsto S_{jj}(\mathbf{k})$, $j=1, \dots, 4$, and cross-spectral density functions $\mathbf{k} \mapsto S_{jk}(\mathbf{k})$, $j \neq k=1, \dots, 4$, of the processes $(X_j(\mathbf{x}), \mathbf{x} \in \mathbb{R}^2)$. For thick shells, they are given by

$$\begin{aligned}
\sigma_{TT}(\mathbf{x}, \mathbf{k}, \mathbf{k}') &= \frac{\pi}{2} \lambda_T^2(\mathbf{x}, \mathbf{k}) [S_{11}(\mathbf{k} - \mathbf{k}') + (\hat{\mathbf{k}} \cdot \hat{\mathbf{k}}')^2 \\
&\times S_{44}(\mathbf{k} - \mathbf{k}') + 2(\hat{\mathbf{k}} \cdot \hat{\mathbf{k}}') \text{Re } S_{14}(\mathbf{k} - \mathbf{k}')], \tag{13}
\end{aligned}$$

and

$$\begin{aligned}
\sigma_{PP}(\mathbf{x}, \mathbf{k}, \mathbf{k}') &= \frac{\pi}{2} \lambda_P^2(\mathbf{x}, \mathbf{k}) \left[(\hat{\mathbf{k}} \cdot \hat{\mathbf{k}}')^2 S_{11}(\mathbf{k} - \mathbf{k}') \right. \\
&+ \left(1 - 2 \frac{c_S^2}{c_P^2} \right)^2 S_{22}(\mathbf{k} - \mathbf{k}') + 4 \frac{c_S^4}{c_P^4} (\hat{\mathbf{k}} \cdot \hat{\mathbf{k}}')^4 \\
&\times S_{33}(\mathbf{k} - \mathbf{k}') + 2 \left(1 - 2 \frac{c_S^2}{c_P^2} \right) (\hat{\mathbf{k}} \cdot \hat{\mathbf{k}}') \\
&\times \text{Re } S_{12}(\mathbf{k} - \mathbf{k}') + 4 \frac{c_S^2}{c_P^2} (\hat{\mathbf{k}} \cdot \hat{\mathbf{k}}')^3 \\
&\times \text{Re } S_{13}(\mathbf{k} - \mathbf{k}') + 4 \frac{c_S^2}{c_P^2} \left(1 - 2 \frac{c_S^2}{c_P^2} \right) \\
&\times (\hat{\mathbf{k}} \cdot \hat{\mathbf{k}}')^2 \text{Re } S_{23}(\mathbf{k} - \mathbf{k}') \left. \right], \tag{14}
\end{aligned}$$

$$\begin{aligned}
\sigma_{SS}(\mathbf{x}, \mathbf{k}, \mathbf{k}') &= \frac{\pi}{2} \lambda_S^2(\mathbf{x}, \mathbf{k}) \left[(\hat{\mathbf{k}} \cdot \hat{\mathbf{k}}')^2 S_{11}(\mathbf{k} - \mathbf{k}') \right. \\
&+ (2(\hat{\mathbf{k}} \cdot \hat{\mathbf{k}}')^2 - 1)^2 S_{33}(\mathbf{k} - \mathbf{k}') + 2(\hat{\mathbf{k}} \cdot \hat{\mathbf{k}}') \\
&\times (2(\hat{\mathbf{k}} \cdot \hat{\mathbf{k}}')^2 - 1) \text{Re } S_{13}(\mathbf{k} - \mathbf{k}') \left. \right], \tag{15}
\end{aligned}$$

$$\begin{aligned}
\sigma_{PS}(\mathbf{x}, \mathbf{k}, \mathbf{k}') &= \frac{\pi}{2} (1 - (\hat{\mathbf{k}} \cdot \hat{\mathbf{k}}')^2) [\lambda_P^2(\mathbf{x}, \mathbf{k}) S_{11}(\mathbf{k} - \mathbf{k}') \\
&+ 4\lambda_S^2(\mathbf{x}, \mathbf{k}) (\hat{\mathbf{k}} \cdot \hat{\mathbf{k}}')^2 S_{33}(\mathbf{k} - \mathbf{k}') \\
&+ 4\lambda_P(\mathbf{x}, \mathbf{k}) \lambda_S(\mathbf{x}, \mathbf{k}) (\hat{\mathbf{k}} \cdot \hat{\mathbf{k}}') \text{Re } S_{13}(\mathbf{k} - \mathbf{k}')], \tag{16}
\end{aligned}$$

and $\sigma_{SP}(\mathbf{x}, \mathbf{k}, \mathbf{k}') = \sigma_{PS}(\mathbf{x}, \mathbf{k}', \mathbf{k})$. At last, the total extinction coefficients $\Sigma_\alpha(\mathbf{x}, \mathbf{k})$ are given by

$$\begin{aligned}
\Sigma_\alpha(\mathbf{x}, \mathbf{k}) &= \sum_{\beta=P,S} \Sigma_{\alpha\beta}(\mathbf{x}, \mathbf{k}) + \eta_\alpha(\mathbf{x}), \quad \alpha = P, S, \\
\Sigma_T(\mathbf{x}, \mathbf{k}) &= \Sigma_{TT}(\mathbf{x}, \mathbf{k}) + \eta_T(\mathbf{x}), \tag{17}
\end{aligned}$$

where $0 < \eta_\alpha < +\infty$ are intrinsic attenuation coefficients, and the total scattering cross sections $\Sigma_{\alpha\beta}$, $\alpha, \beta = T, P, S$, are computed by

$$\Sigma_{\alpha\beta}(\mathbf{x}, \mathbf{k}) = \int_{\mathbb{R}^2} \sigma_{\alpha\beta}(\mathbf{x}, \mathbf{k}, \mathbf{k}') \delta(\lambda_\alpha(\mathbf{x}, \mathbf{k}) - \lambda_\beta(\mathbf{x}, \mathbf{k}')) d\mathbf{k}'. \tag{18}$$

The above expressions of scattering cross sections are derived from a model of random media where fluctuations are seen as potentially continuous; however, their particularization to discrete scatterers is straightforward, using the expression of the scattering cross section of a single scatterer multiplied by the density of scatterers, as in, e.g., Refs. 38–40. Inclusions in the form of random localized boundary impedances may also be considered.^{41,42} Clearly, it is first seen from these results that the high-frequency energy density evolution is independent of the shell curvature tensor $\mathbf{B}$. Therefore, all equations above apply to plates as well. Second, the pure transverse shear energy decouples from the compressional and shear energies because Eq. (9) and Eqs. (10) and (11) are uncoupled. In addition, the compressional and shear energies are coupled entrywise only, that is, their different entries do not mix. Thus only the diagonals of Eqs. (10) and (11) need be retained for the computation of the overall energy density (8). Third, Eqs. (9)–(11) with the definition of the Poisson bracket indicate that the energy densities are transported at speeds equal to the group velocities:

$$\mathbf{c}_{g,\alpha}(\mathbf{x}, \mathbf{k}) := \nabla_{\mathbf{k}} \lambda_\alpha(\mathbf{x}, \mathbf{k}) = c_\alpha(\mathbf{x}) \hat{\mathbf{k}}. \tag{19}$$

Their amplitudes $|\mathbf{c}_{g,\alpha}|$ are equal to the corresponding phase velocities because plate/shell waves are nondispersive in the high-frequency setting expounded above (the c_α 's are independent of the frequency in the limit $k_E/k_P \rightarrow 0$, where k_E is the bending wavenumber for Kirchhoff–Love thin plate kinematics, and k_P is the bending wavenumber for Mindlin–Reissner–Uflyand thick plate kinematics considered here). This model differs from a Lamb model, where a plate is defined as a three-dimensional elastic medium bounded by two traction-free surfaces at $z = \pm h/2$. Lamb waves are dispersive, and the group velocities of the associated energy densities are clearly different from the phase velocities. This situation has been considered in, e.g., Ref. 38, and would apply to short wavelengths Λ with respect to the plate thickness h . However, in the very limit $\Lambda/h \rightarrow 0$, Lamb modes also become nondispersive.

B. Beams

The constitutive equations and equations of motion for a thick curved beam in the plane $\mathbb{R}^2$ are easily deduced from the previous expressions (2) and (3) for shells. The beam

centerline $\mathcal{L}$ is parametrized by its curvilinear abscissa $x \in \mathbb{R}$, and its tangent motion, deflection, and bending rotation are denoted by $u^\epsilon(x)$, $w^\epsilon(x)$, and $\theta^\epsilon(x)$, respectively. The local tangent and normal vectors to $\mathcal{L}$ are denoted by $\mathbf{e}$ and $\hat{\mathbf{n}}$, respectively. The overall displacement field of the beam in the local basis $(\mathbf{e}, \hat{\mathbf{n}})$ thus writes

$$\mathbf{U}^\epsilon(x) = (u^\epsilon(x) - z\theta^\epsilon(x))\mathbf{e}(x) + w^\epsilon(x)\hat{\mathbf{n}}(x),$$

which is nothing but the (Timoshenko) kinematic assumption defining a beam in a plane, up to the first order in the radial coordinate z . Note that torsional motions are not considered here, but the following analysis can be generalized to beams in $\mathbb{R}^3$ to account for them, as well as for the coupled bending waves along both principal normal directions. Integrating the constitutive equation over the beam cross section yields

$$N^\epsilon = E^\epsilon S \left(\partial_x u^\epsilon + \frac{w^\epsilon}{R} \right), \quad T^\epsilon = \mu_r^\epsilon S \left(\partial_x w^\epsilon - \theta^\epsilon + \frac{u^\epsilon}{R} \right), \quad (20)$$

$$M^\epsilon = E^\epsilon I \partial_x \theta^\epsilon,$$

for the compressional force N^ϵ , shear force T^ϵ , and bending moment M^ϵ ; S and I are the beam cross-section area and inertia, respectively, and $R(x)$ is its radius of curvature. The equations of motion are

$$\rho^\epsilon \partial_t^2 u^\epsilon = \partial_x N^\epsilon - \frac{T^\epsilon}{R}, \quad \rho^\epsilon \partial_t^2 w^\epsilon = \partial_x T^\epsilon - \frac{N^\epsilon}{R}, \quad (21)$$

$$\frac{\rho^\epsilon I}{S} \partial_t^2 \theta^\epsilon = \partial_x M^\epsilon + T^\epsilon.$$

Now assume as for shells that the elastic parameters are

$$\begin{aligned} \rho^\epsilon(x) &= \underline{\rho}(x) \left[1 + \sqrt{\epsilon} X_1 \left(\frac{x}{\epsilon} \right) \right], \\ \frac{1}{E^\epsilon(x)} &= \frac{1}{\underline{E}(x)} \left[1 + \sqrt{\epsilon} X_2 \left(\frac{x}{\epsilon} \right) \right], \\ \frac{1}{\mu_r^\epsilon(x)} &= \frac{1}{\underline{\mu}_r(x)} \left[1 + \sqrt{\epsilon} X_3 \left(\frac{x}{\epsilon} \right) \right], \end{aligned} \quad (22)$$

where $(X_j(x), x \in \mathbb{R})$, $j=1,2,3$, are three real-valued statistically homogeneous and centered second-ordered stochastic processes with correlation lengths comparable to the wavelength in the medium, and $\underline{\rho}(x) = \mathbb{E}\{\rho^\epsilon(x)\}$, $\underline{E}(x) = \mathbb{E}\{E^\epsilon(x)^{-1}\}^{-1}$, and $\underline{\mu}_r(x) = \mathbb{E}\{\mu_r^\epsilon(x)^{-1}\}^{-1}$ are the averaged macroscopic density, Young's modulus, and reduced shear modulus, respectively. As for shells, the perturbation amplitude scaling above is chosen in order to evade localization while the energy propagation pattern is still significantly perturbed by the heterogeneities. Of course localization in such a one-dimensional disordered medium is unavoidable in the end; however, the localization length is expected to be much larger than the observation distances imposed by this setting; for example, its *minimum* value is of order ϵ^{-1} in the framework of Ref. 43. Then there exist positive scalars w_T and w_P such that the elastic energy density of the beam in the high-frequency limit $\epsilon \rightarrow 0$ is

$$\lim_{\epsilon \rightarrow 0} \mathcal{E}^\epsilon(x, t) = \sum_{\alpha=T,P} \int_{\mathbb{R}} w_\alpha(x, k, t) dk. \quad (23)$$

They satisfy the RTEs:

$$\begin{aligned} \partial_t w_\alpha + \{\lambda_\alpha, w_\alpha\} &= \int_{\mathbb{R}} \sigma_{\alpha\alpha}(x, k, k') w_\alpha(x, k', t) \delta(\lambda_\alpha(x, k) \\ &\quad - \lambda_\alpha(x, k')) dk' - \Sigma_\alpha(x, k) w_\alpha(x, k, t), \end{aligned} \quad (24)$$

where $\lambda_\alpha(x, k) = c_\alpha(x)|k|$ for either $\alpha=T$ or $\alpha=P$ with

$$c_T = \sqrt{\frac{\mu_r S}{\rho}}, \quad c_P = \sqrt{\frac{ES}{\rho}}.$$

Again, w_T and w_P are interpreted as phase-space energy densities in $T^*\mathbb{R} = \mathbb{R}_x \times \mathbb{R}_k$ associated with pure transverse waves polarized in the normal direction, and coupled bending-compressional waves polarized in the propagation direction $\hat{k} = \text{sign}(k) = \pm 1$. The differential scattering cross sections $\sigma_{\alpha\alpha}$ in Eq. (24) are given by

$$\begin{aligned} \sigma_{TT}(x, k, k') &= \frac{\pi}{2} \lambda_T^2(x, k) [S_{11}(k - k') + S_{33}(k - k') \\ &\quad + 2 \text{sign}(k) \text{sign}(k') \text{Re } S_{13}(k - k')], \\ \sigma_{PP}(x, k, k') &= \frac{\pi}{2} \lambda_P^2(x, k) [S_{11}(k - k') + S_{22}(k - k') \\ &\quad + 2 \text{sign}(k) \text{sign}(k') \text{Re } S_{12}(k - k')], \end{aligned}$$

and the extinction coefficients Σ_α are as for shells:

$$\begin{aligned} \Sigma_\alpha(x, k) &= \eta_\alpha(x) + \int_{\mathbb{R}} \sigma_{\alpha\alpha}(x, k, k') \delta(\lambda_\alpha(x, k) \\ &\quad - \lambda_\alpha(x, k')) dk', \quad \alpha = T, P, \end{aligned}$$

where $0 < \eta_\alpha < +\infty$. Note that all transport characteristics above are independent of the radius of curvature of the beam. The group velocities $\pm c_T$ and $\pm c_P$ are equal to the phase velocities because waves in thick beams are nondispersive in the high-frequency limit $k_E/k_P \rightarrow 0$, the ratio of bending wavenumbers for Euler–Bernoulli thin beam and Timoshenko thick beam kinematics, respectively.

III. MULTIGROUP RADIATIVE TRANSFER EQUATIONS

In this section, the characteristic features of the coupled RTEs satisfied by the phase-space energy densities are discussed. The crucial role played by scattering on random heterogeneities as described by the integral operators in Eqs. (9)–(11) or Eq. (24) is emphasized. The directional part of these scattering processes is analyzed more in detail in Sec. III B below.

A. Transport versus diffusion

Let d be the physical space dimension, typically $d=2$ for a plate or a shell and $d=1$ for a beam. The streaming operator in phase space $T^*\mathbb{R}^d$ corresponding to the Poisson bracket on the left-hand-side in Eqs. (9)–(11) for plates and shells or

Eq. (24) for beams describes the propagation of the phase-space energy density associated with a given mode α at a velocity $|\mathbf{c}_{g,\alpha}|$ along rays which are the bicharacteristics of the Hamiltonian $\lambda_\alpha(\mathbf{x}, \mathbf{k})$. The latter are defined as the functions $t \mapsto \mathbf{x}_\alpha(t)$ and $t \mapsto \mathbf{k}_\alpha(t)$ solving the ordinary differential equations:

$$\dot{\mathbf{x}}_\alpha(t) = \nabla_{\mathbf{k}} \lambda_\alpha(\mathbf{x}_\alpha(t), \mathbf{k}_\alpha(t)),$$

$$\dot{\mathbf{k}}_\alpha(t) = -\nabla_{\mathbf{x}} \lambda_\alpha(\mathbf{x}_\alpha(t), \mathbf{k}_\alpha(t)),$$

in $T^*\mathbb{R}^d \setminus \{(\mathbf{x}, \mathbf{k}); \mathbf{k} = 0\}$, such that

$$\partial_t w + \{\lambda_\alpha, w\} \equiv \frac{dw}{dt}.$$

Then the Hamiltonian λ_α is constant along its bicharacteristics.

The scattering or linear Boltzmann's operators corresponding to the right-hand sides in Eqs. (9)–(11) or Eq. (24) describe the various interaction processes undergone by a ray $[\mathbf{x}(t), \mathbf{k}(t)]$ in phase space $T^*\mathbb{R}^d$ transporting the energy density w_α : (a) collisions on random heterogeneities which convert all incident rays in the directions $\hat{\mathbf{k}}'$ and modes β into $\hat{\mathbf{k}}$ and α , respectively, at the rate $\sigma_{\alpha\beta}(\mathbf{x}, \mathbf{k}, \mathbf{k}')$ at the position $\mathbf{x}$, balanced by the reciprocal conversion of the actual direction $\hat{\mathbf{k}}$ and mode α to all other directions and modes at the rate $\sum_\beta \Sigma_{\alpha\beta}(\mathbf{x}, \mathbf{k})$; (b) attenuation at the rate $\eta_\alpha(\mathbf{x})$. The scattering process (a) is conservative and characterized by a so-called scattering mean free path ℓ which is defined for a particular energy mode α by

$$\ell_\alpha(\mathbf{x}, \mathbf{k}) = \frac{|\mathbf{c}_{g,\alpha}(\mathbf{x}, \mathbf{k})|}{\sum_\alpha(\mathbf{x}, \mathbf{k})}. \quad (25)$$

ℓ_α corresponds to the mean distance traveled by an energy ray in the mode α before its direction $\hat{\mathbf{k}}$ is significantly altered by scattering on the heterogeneities. Derivation of the RTEs (9)–(11) or (24) is carried out in the limit $\epsilon \equiv \Lambda/L \rightarrow 0$, where Λ is a typical wavelength and $L \equiv c_\alpha t$ is the distance at which propagation of the mode α is observed. This transport regime remains applicable as long as $\Lambda \ll \ell_\alpha$, while it does not require short wavelength compared with the size of heterogeneities. A diffusion regime sets up when (i) attenuations in the medium are small, (ii) scattering is important, and (iii) one considers propagation distances/times much larger than the scattering mean free paths/times, or equivalently multiply scattered energy rays: $c_\alpha t \gg \ell_\alpha$. Scattering operators play a crucial role in the emergence of this regime in that they mix all directions of all existing energy modes. Therefore, the analysis of the diffusion limit relies on an explicit identification of their directional part; this is done in the next section. For that purpose, both Eq. (9) or Eq. (24) and the diagonals of Eqs. (10) and (11) [in view of estimating the energy density by Eq. (8)] are recast into the following system of multigroup transport equations:

$$\begin{aligned} \partial_t w_\alpha + \{\lambda_\alpha, w_\alpha\} &= \sum_{\beta=1}^M \int_{\mathbb{R}^d} \sigma_{\alpha\beta}(\mathbf{x}, \mathbf{k}, \mathbf{k}') w_\beta(\mathbf{x}, \mathbf{k}', t) \delta(\lambda_\alpha(\mathbf{x}, \mathbf{k}) \\ &\quad - \lambda_\beta(\mathbf{x}, \mathbf{k}')) d\mathbf{k}' - \Sigma_\alpha(\mathbf{x}, \mathbf{k}) w_\alpha(\mathbf{x}, \mathbf{k}, t), \\ \alpha &= 1, 2, \dots, M. \end{aligned} \quad (26)$$

The remaining of the paper considers this generic form for an arbitrary *finite* number m of different energy modes, each having an order of multiplicity R_α such that $\sum_{\alpha=1}^m R_\alpha = M$. The particular case of shells is recovered from this model provided that one defines $m = M = R_T = 1$ for Eq. (9) or $m = 2$ and $R_P = R_S = 2$ such that $M = 4$ for Eqs. (10) and (11); for curved beams, $m = M = R_\alpha = 1$ for either $\alpha = T$ or P (of course the interesting case is $m > 1$). Once the solutions $\{w_\alpha\}_{\alpha=1, \dots, M}$ of Eq. (26) with known initial conditions have been recovered, the high-frequency energy in the medium is obtained as

$$\lim_{\epsilon \rightarrow 0} \mathcal{E}^\epsilon(t) := \mathcal{E}(t) = \sum_{\alpha=1}^M \int_{T^*\mathbb{R}^d} w_\alpha(\mathbf{x}, \mathbf{k}, t) d\mathbf{k} d\mathbf{x},$$

while the associated high-frequency power flow vector is

$$\lim_{\epsilon \rightarrow 0} \mathbf{\Pi}^\epsilon(t) := \mathbf{\Pi}(t) = \sum_{\alpha=1}^M \int_{T^*\mathbb{R}^d} \mathbf{c}_{g,\alpha}(\mathbf{x}, \mathbf{k}) w_\alpha(\mathbf{x}, \mathbf{k}, t) d\mathbf{k} d\mathbf{x},$$

recalling the definition (19) of the group velocities $\mathbf{c}_{g,\alpha}$.

To conclude on this section, we note that if all scattering cross sections are zero, that is, if parameters ρ^ϵ , E^ϵ , and ν^ϵ in Sec. II vary at the macroscopic scale $O(1)$ only and not at the microscopic scale $O(\epsilon)$, then the scattering mean free paths (25) are infinite and no diffusion limit arises. This is the regime of pure transport of high-frequency waves, where by Eqs. (9)–(11) for shells or Eq. (24) for plates, energy rays for distinct modes evolve without ever mixing, even if the background medium is heterogeneous at the macroscopic scale. However, a strong mixing of directions and polarizations (modes) may be obtained by the reflections and transmissions of the energy rays at boundaries, interfaces, or junctions between different substructures. One may also define finite scattering mean paths in this situation, which is considered in room acoustics or in the SEA of engineering structures (see Ref. 20). Boundary conditions will *not* be considered in this paper. In Ref. 31, for example, the authors have derived diffusion equations relevant to either smooth or rough boundaries corresponding to specularlike or diffusive-like reflections, respectively, in a general framework.

B. Renormalization of the scattering operators

Let us introduce the spectral density matrix $\mathbf{S}(\mathbf{k})$ of parameters random fluctuations, defined by $[\mathbf{S}(\mathbf{k})]_{jk} = S_{jk}(\mathbf{k})$, the power ($j = k$) or cross- ($j \neq k$) spectral density function of random fluctuations ($X_j(\mathbf{x})$, $\mathbf{x} \in \mathbb{R}^d$) and ($X_k(\mathbf{x})$, $\mathbf{x} \in \mathbb{R}^d$); it is a four-by-four complex matrix for shells and plates or a three-by-three complex matrix for beams. A spectral density matrix is always positive, Hermitian, and satisfies $\mathbf{S}(-\mathbf{k}) = \mathbf{S}(\mathbf{k}) = \mathbf{S}(\mathbf{k})^T$, where $\mathbf{A}^T$ stands for the matrix transpose and $\bar{z}$ stands for the complex conjugate. Thus $\mathbf{k} \mapsto \text{Re } S_{jk}(\mathbf{k})$ are even functions and $\mathbf{k} \mapsto S_{jj}(\mathbf{k})$ are in addition real. Differential scattering cross sections $\sigma_{\alpha\beta}(\mathbf{x}, \mathbf{k}, \mathbf{k}')$

in Eq. (26) are completely determined by $\mathbf{S}(\mathbf{k}-\mathbf{k}')$, thus they are always non-negative¹¹ owing to the positivity of $\mathbf{S}$. Its symmetries imply that they also satisfy the reciprocity relation:

$$\sigma_{\alpha\beta}(\mathbf{x}, \mathbf{k}, \mathbf{k}') \delta(\lambda_\alpha(\mathbf{x}, \mathbf{k}) - \lambda_\beta(\mathbf{x}, \mathbf{k}')) = \sigma_{\beta\alpha}(\mathbf{x}, \mathbf{k}', \mathbf{k}) \delta(\lambda_\beta(\mathbf{x}, \mathbf{k}') - \lambda_\alpha(\mathbf{x}, \mathbf{k})).$$

as long as there is scattering between modes α and β . If the fluctuations $(X_f(\mathbf{x}), \mathbf{x} \in \mathbb{R}^d)$ are statistically isotropic, then S_{jj} and $\text{Re } S_{jk}$ depend on $|\mathbf{k}|$ solely, and the differential scattering cross sections depend on $|\mathbf{k}|$ and $\hat{\mathbf{k}} \cdot \hat{\mathbf{k}}'$ solely because $|\mathbf{k}-\mathbf{k}'| = |\mathbf{k}|(1 + (c_\alpha^2/c_\beta^2) - 2(c_\alpha/c_\beta)\hat{\mathbf{k}} \cdot \hat{\mathbf{k}}')^{1/2}$ whenever $\lambda_\alpha(\mathbf{x}, \mathbf{k}) = \lambda_\beta(\mathbf{x}, \mathbf{k}')$. They are now written as

$$\sigma_{\alpha\beta}(\mathbf{x}, \mathbf{k}, \mathbf{k}') \delta(\lambda_\alpha(\mathbf{x}, \mathbf{k}) - \lambda_\beta(\mathbf{x}, \mathbf{k}')) = \frac{c_\beta}{|\mathbf{k}'|^{d-1}} \tilde{\sigma}_{\alpha\beta}(\mathbf{x}, |\mathbf{k}|, |\mathbf{k}'|, \hat{\mathbf{k}} \cdot \hat{\mathbf{k}}') \delta(\lambda_\alpha(\mathbf{x}, \mathbf{k}) - \lambda_\beta(\mathbf{x}, \mathbf{k}'))$$

such that

$$\tilde{\sigma}_{\alpha\beta}\left(\mathbf{x}, |\mathbf{k}|, \frac{c_\alpha}{c_\beta}|\mathbf{k}|, \hat{\mathbf{k}} \cdot \hat{\mathbf{k}}'\right) = \left(\frac{c_\alpha}{c_\beta}\right)^d \tilde{\sigma}_{\beta\alpha}\left(\mathbf{x}, \frac{c_\alpha}{c_\beta}|\mathbf{k}|, |\mathbf{k}|, \hat{\mathbf{k}} \cdot \hat{\mathbf{k}}'\right) \quad (27)$$

in order to satisfy the reciprocity property of the $\sigma_{\alpha\beta}$'s. This last relation is known as the detailed balance principle in the neutron transport literature. Then,

$$\begin{aligned} \Sigma_{\alpha\beta}(\mathbf{x}, \mathbf{k}) &= \tilde{\Sigma}_{\alpha\beta}(\mathbf{x}, |\mathbf{k}|) \\ &= \int_{S^{d-1}} \tilde{\sigma}_{\alpha\beta}\left(\mathbf{x}, |\mathbf{k}|, \frac{c_\alpha}{c_\beta}|\mathbf{k}|, \hat{\mathbf{k}} \cdot \hat{\mathbf{k}}'\right) d\Omega(\hat{\mathbf{k}}') \\ &= \left(\frac{c_\alpha}{c_\beta}\right)^d \tilde{\Sigma}_{\beta\alpha}\left(\mathbf{x}, \frac{c_\alpha}{c_\beta}|\mathbf{k}|\right), \end{aligned} \quad (28)$$

where S^{d-1} is the unit sphere of $\mathbb{R}^d$ and $d\Omega$ is its Euclidean measure. As $d\mathbf{k} = r^{d-1} dr d\Omega(\hat{\mathbf{k}})$ with $r = |\mathbf{k}|$, Eq. (26) yields

$$\begin{aligned} \partial_t w_\alpha + \{\lambda_\alpha, w_\alpha\} &= \sum_{\beta=1}^M \int_{S^{d-1}} \tilde{\sigma}_{\alpha\beta}\left(\mathbf{x}, |\mathbf{k}|, \frac{c_\alpha}{c_\beta}|\mathbf{k}|, \hat{\mathbf{k}} \cdot \hat{\mathbf{k}}'\right) \\ &\quad \times w_\beta\left(\mathbf{x}, \frac{c_\alpha}{c_\beta}|\mathbf{k}| \hat{\mathbf{k}}', t\right) d\Omega(\hat{\mathbf{k}}') \\ &\quad - \tilde{\Sigma}_\alpha(\mathbf{x}, |\mathbf{k}|) w_\alpha(\mathbf{x}, |\mathbf{k}| \hat{\mathbf{k}}, t), \quad \alpha = 1, 2, \dots, M, \end{aligned} \quad (29)$$

where $\tilde{\Sigma}_\alpha = \sum_{\beta=1}^M \tilde{\Sigma}_{\alpha\beta} + \eta_\alpha$. The last step of this analysis is to project the streaming operators onto the unit phase space $S^*\mathbb{R}^d \equiv \mathbb{R}_x^d \times S_{\mathbf{k}}^{d-1}$ (the unit cosphere bundle of $\mathbb{R}^d$). The multigroup transport equations for the phase-space energy densities $w_\alpha(\mathbf{x}, \mathbf{k}, t)$ are conveniently written in the form of a coupled system for the so-called specific intensities $a_\alpha(\mathbf{x}, \omega, \hat{\mathbf{k}}, t)$, or radiances in the optical literature,³⁵ defined here by

$$a_\alpha(\mathbf{x}, \omega, \hat{\mathbf{k}}, t) = w_\alpha\left(\mathbf{x}, \frac{\omega}{c_\alpha(\mathbf{x})} \hat{\mathbf{k}}, t\right). \quad (30)$$

In Eq. (29), or in Eq. (26) above, $\omega = \lambda_\alpha(\mathbf{x}, \mathbf{k}) = c_\alpha(\mathbf{x})|\mathbf{k}|$ is a fixed parameter along the rays, and the projection of the usual gradient $\nabla_{\mathbf{k}}$ in $\mathbb{R}^d$ onto S^{d-1} is $(\mathbf{I}_d - \hat{\mathbf{k}} \otimes \hat{\mathbf{k}}) \nabla_{\hat{\mathbf{k}}}$, where $\mathbf{I}_d$ is the $d \times d$ identity matrix. Specific intensities $\{a_\alpha\}_{\alpha=1, \dots, M}$ are then solutions of the multigroup transport equations:

$$\begin{aligned} \partial_t a_\alpha + c_\alpha(\mathbf{x}) \hat{\mathbf{k}} \cdot \nabla_{\mathbf{x}} a_\alpha - \nabla_{\mathbf{x}} c_\alpha(\mathbf{x}) \cdot (\mathbf{I}_d - \hat{\mathbf{k}} \otimes \hat{\mathbf{k}}) \nabla_{\hat{\mathbf{k}}} a_\alpha \\ + \eta_\alpha(\mathbf{x}) a_\alpha = \sum_{\beta=1}^M \int_{S^{d-1}} \tilde{\sigma}_{\alpha\beta}\left(\mathbf{x}, \frac{\omega}{c_\alpha}, \frac{\omega}{c_\beta}, \hat{\mathbf{k}} \cdot \hat{\mathbf{k}}'\right) \\ \times (a_\beta(\mathbf{x}, \omega, \hat{\mathbf{k}}', t) - a_\alpha(\mathbf{x}, \omega, \hat{\mathbf{k}}, t)) d\Omega(\hat{\mathbf{k}}'). \end{aligned} \quad (31)$$

The high-frequency energy of the medium is now given by

$$\begin{aligned} \mathcal{E}(t) &= \sum_{\alpha=1}^M \int_{S^*\mathbb{R}^d} c_\alpha^{-d}(\mathbf{x}) \\ &\quad \times \left(\int_0^{+\infty} \omega^{d-1} a_\alpha(\mathbf{x}, \omega, \hat{\mathbf{k}}, t) d\omega \right) d\Omega(\hat{\mathbf{k}}) d\mathbf{x}, \end{aligned} \quad (32)$$

and the power flow vector by

$$\begin{aligned} \Pi(t) &= \sum_{\alpha=1}^M \int_{S^*\mathbb{R}^d} c_\alpha^{-d}(\mathbf{x}) \mathbf{c}_{g,\alpha}(\mathbf{x}, \mathbf{k}) \\ &\quad \times \left(\int_0^{+\infty} \omega^{d-1} a_\alpha(\mathbf{x}, \omega, \hat{\mathbf{k}}, t) d\omega \right) d\Omega(\hat{\mathbf{k}}) d\mathbf{x}. \end{aligned} \quad (33)$$

It should be noted that the a_α 's, as well as the w_α 's, all have their values in $\mathbb{R}_+$. Also the group velocities $\mathbf{c}_{g,\alpha}(\mathbf{x}, \mathbf{k}) = c_\alpha(\mathbf{x}) \hat{\mathbf{k}}$ in the model equation (31) have amplitudes independent of the wavenumber and equal to the phase velocities c_α , as in Sec. II A or II B. The assumption of statistical isotropy invoked above is technically essential for the derivation which follows. Diffusion approximations can also be constructed for statistically anisotropic media (called anisomeric random media) by a rather identical approach, see, e.g., Ref. 6, invoking, however, different technical arguments. This issue will not be pursued in the subsequent analysis.

IV. DIFFUSION REGIME IN THE LIMIT OF SMALL SCATTERING MEAN FREE PATHS

In this section, one introduces the small parameter $\delta = \ell_\alpha/L$ and let $\delta \rightarrow 0$ to obtain the diffusion limit by an asymptotic expansion method, the so-called Hilbert expansion method. The derivation is based on the algebraic properties of the conservative part of the scattering operators recalled below. Some important features of the diffusive regime that can be obtained from the analysis are also outlined: depolarization of the waves, equipartition of their energy densities, and asymptotic Fick's law.

A. Properties of the collision operators

We first give the main properties of the linear collision operators on the right-hand side of Eq. (31). They correspond to the conservative part of the scattering operators on the unit sphere S^{d-1} . Define the average over S^{d-1} by

$$\langle f \rangle = \frac{1}{\Omega_d} \int_{S^{d-1}} f(\hat{\mathbf{k}}) d\Omega(\hat{\mathbf{k}}), \quad \Omega_d = \int_{S^{d-1}} d\Omega(\hat{\mathbf{k}}).$$

For $\mathbf{x} \in \mathcal{O} \subseteq \mathbb{R}^d$ and $\omega > 0$, let

$$\mathcal{K}_\omega f(\hat{\mathbf{k}}) = \frac{1}{\Omega_d} \int_{S^{d-1}} \boldsymbol{\sigma}(\mathbf{x}, \omega, \hat{\mathbf{k}} \cdot \hat{\mathbf{k}}') f(\hat{\mathbf{k}}') d\Omega(\hat{\mathbf{k}}'),$$

$$\mathcal{Q}_\omega f(\hat{\mathbf{k}}) = \Sigma_\omega f(\hat{\mathbf{k}}) - \mathcal{K}_\omega f(\hat{\mathbf{k}}),$$

where

$$\Sigma_\omega = \frac{1}{\Omega_d} \tilde{\Sigma} \mathbf{C}^d, \quad \boldsymbol{\sigma} = \tilde{\boldsymbol{\sigma}} \mathbf{C}^d, \quad \mathbf{C} = \text{diag}(c_\alpha)_{\alpha=1, \dots, M},$$

$$\tilde{\Sigma} = \text{diag} \left(\sum_{\beta=1}^M \tilde{\Sigma}_{\alpha\beta} \right)_{\alpha=1, \dots, M},$$

and $\tilde{\boldsymbol{\sigma}}$ is the $M \times M$ matrix of which elements are the scattering cross sections $\tilde{\sigma}_{\alpha\beta}$ such that Eq. (27) holds

$$[\tilde{\boldsymbol{\sigma}}(\mathbf{x}, \omega, \hat{\mathbf{k}} \cdot \hat{\mathbf{k}}')]_{\alpha\beta} = \tilde{\sigma}_{\alpha\beta} \left(\mathbf{x}, \frac{\omega}{c_\alpha}, \frac{\omega}{c_\beta}, \hat{\mathbf{k}} \cdot \hat{\mathbf{k}}' \right).$$

Then $\mathcal{Q}_\omega$ is self-adjoint on $H = (L^2(S^{d-1}, d\Omega))^M$; if we assume that $\boldsymbol{\sigma}$ is such that

$$\sup_{\mathbf{x} \in \mathcal{O}, \omega > 0} \int \int_{S^{d-1} \times S^{d-1}} \|\boldsymbol{\sigma}(\mathbf{x}, \omega, \hat{\mathbf{k}} \cdot \hat{\mathbf{k}}')\|_2^2 d\Omega(\hat{\mathbf{k}}) d\Omega(\hat{\mathbf{k}}') < +\infty,$$

with the entrywise matrix norm $\|\mathbf{A}\|_p = (\sum_{\alpha, \beta=1}^M |A_{\alpha\beta}|^p)^{1/p}$, then $\mathcal{K}_\omega$ is a Hilbert–Schmidt operator on H . Owing to the fact that the scattering cross sections are non-negative functions, such that $\exists \alpha, \tilde{\sigma}_{\alpha\beta} > 0$ for each $\beta = 1, 2, \dots, M$ [otherwise the RTEs are split in subgroups, and diffusion limits are reached for each group separately, as for the two groups of Eqs. (10) and (11) and Eq. (9) above, respectively], one can show in addition that the null space of $\mathcal{Q}_\omega$ is

$$N(\mathcal{Q}_\omega) = \{\phi \mathbf{d}; \phi = \langle \phi \rangle\}, \quad (34)$$

with $\mathbf{d} = \mathbf{C}^{-d} \mathbf{u}$, $\mathbf{u} = (1, 1, \dots, 1)^T$ in $\mathbb{R}^M$, and ϕ scalar functions independent of $\hat{\mathbf{k}}$. Indeed, $\mathcal{K}_\omega$ is compact and strongly positive on the cone $K = \{f \in H; f_\alpha \geq 0, \alpha = 1, 2, \dots, M\}$, and one observes that $\mathbf{d}$ is a positive eigenvector of $\mathcal{K}_\omega$. Thus by the Krein–Rutman theorem,⁴⁴ it is the sole positive eigenvector associated with the simple eigenvalue $\rho(\mathcal{K}_\omega)$, its spectral radius, that is, $N(\mathcal{Q}_\omega)$ is the one-dimensional subspace of H spanned by $\mathbf{d}$. Now by Fredholm alternative, the equation $\mathcal{Q}_\omega f = g$ in $N(\mathcal{Q}_\omega)^\perp = R(\mathcal{Q}_\omega)$ has a unique solution f^* , which satisfies $(f^*, \phi \mathbf{d})_H = \phi \mathbf{d} \cdot \langle f^* \rangle = 0$ for any constant function ϕ .

B. Asymptotic analysis

Recalling conditions (i)–(iii) of Sec. III A for the emergence of a diffusive regime, the system of Eq. (31) is rescaled as

$$\begin{aligned} & \delta \partial_t a_\alpha^\delta + c_\alpha \hat{\mathbf{k}} \cdot \nabla_{\mathbf{x}} a_\alpha^\delta - \nabla_{\mathbf{x}} c_\alpha \cdot (\mathbf{I}_d - \hat{\mathbf{k}} \otimes \hat{\mathbf{k}}) \nabla_{\hat{\mathbf{k}}} a_\alpha^\delta + \delta \eta_\alpha a_\alpha^\delta \\ &= \sum_{\beta=1}^M \int_{S^{d-1}} \delta^{-1} \tilde{\sigma}_{\alpha\beta} \left(\mathbf{x}, \frac{\omega}{c_\alpha}, \frac{\omega}{c_\beta}, \hat{\mathbf{k}} \cdot \hat{\mathbf{k}}' \right) (a_\beta^\delta(\mathbf{x}, \omega, \hat{\mathbf{k}}', t) \\ & \quad - a_\alpha^\delta(\mathbf{x}, \omega, \hat{\mathbf{k}}, t)) d\Omega(\hat{\mathbf{k}}'), \end{aligned} \quad (35)$$

for $\alpha = 1, 2, \dots, M$. This setting corresponds to the near critical case of Ref. 17; however, different scalings may be considered, for example, splitting the collision operator into its elastic (jumps in directions only) and inelastic (jumps in energy modes only) parts and adopt a different scaling for both, as in Ref. 29. It also allows the specific intensities to be effectively observed at large times or distances with respect to mean free times/distances after multiple scattering on small perturbations, diverting potential localization effects. The fact that the latter always prevail at late times in dimension $d=1$ or 2 does not invalidate the following developments; however, the existence of a localization length ξ_d for $d \leq 2$ implies that the diffusive regime holds in an intermediate range between ℓ_α and ξ_d with $\delta^{-1} \ell_\alpha \ll \xi_d$ in accordance with the scaling of Eq. (35). Let $\mathbf{a}^\delta = (a_1^\delta, a_2^\delta, \dots, a_M^\delta)^T$, then it is expanded in power series of δ as

$$\begin{aligned} \mathbf{a}^\delta(\mathbf{x}, \omega, \hat{\mathbf{k}}, t) &= \mathbf{a}^{(0)}(\mathbf{x}, \omega, \hat{\mathbf{k}}, t) + \delta \mathbf{a}^{(1)}(\mathbf{x}, \omega, \hat{\mathbf{k}}, t) \\ & \quad + \delta^2 \mathbf{a}^{(2)}(\mathbf{x}, \omega, \hat{\mathbf{k}}, t) + o(\delta^2) \end{aligned}$$

and plugged into Eq. (35). Balancing order by order, one has to the leading order $O(\delta^0)$:

$$\mathcal{Q}_\omega \mathbf{C}^{-d} \mathbf{a}^{(0)} = \mathbf{0}, \quad (36)$$

which shows that $\mathbf{a}^{(0)}$ does not depend on $\hat{\mathbf{k}}$, because $\mathbf{C}^{-d} \mathbf{a}^{(0)}$ lies in the null space (34) of $\mathcal{Q}_\omega$. Therefore, $\mathbf{C}^{-d} \mathbf{a}^{(0)} = \phi \mathbf{d}$, where $\phi = \langle \phi \rangle$, that is, $\mathbf{a}^{(0)} = \phi \mathbf{u}$: this is depolarization of initially coupled modes, stating that

$$a_\alpha^{(0)}(\mathbf{x}, \omega, t) = a_\beta^{(0)}(\mathbf{x}, \omega, t) := \phi(\mathbf{x}, \omega, t), \quad (37)$$

$$\forall \alpha, \beta = 1, 2, \dots, M,$$

independent of the direction $\hat{\mathbf{k}}$.

To the first order $O(\delta)$, one has

$$-\Omega_d \mathcal{Q}_\omega \mathbf{C}^{-d} \mathbf{a}^{(1)} = (\mathbf{C} \otimes \hat{\mathbf{k}}) \nabla_{\mathbf{x}} \otimes \mathbf{a}^{(0)}. \quad (38)$$

Its general solution is

$$\mathbf{a}^{(1)}(\mathbf{x}, \omega, \hat{\mathbf{k}}, t) = \mathbf{a}^{(11)}(\mathbf{x}, \omega, t) - (\boldsymbol{\Lambda}^* \otimes \hat{\mathbf{k}}) \nabla_{\mathbf{x}} \otimes \mathbf{a}^{(0)}(\mathbf{x}, \omega, t), \quad (39)$$

where $\boldsymbol{\Lambda}^*(\mathbf{x}, \omega)$ is the solution of $\mathbf{C} \otimes \hat{\mathbf{k}} = \Omega_d \mathcal{Q}_\omega (\mathbf{C}^{-d} \boldsymbol{\Lambda}^* \otimes \hat{\mathbf{k}})$, which is uniquely invertible since $\langle \mathbf{C} \otimes \hat{\mathbf{k}} \rangle = \mathbf{0}$. Representation (39) above holds because $\hat{\mathbf{k}} \cdot \mathbf{e}$ is an eigenfunction of the operator $\mathbf{K}_0 f(\hat{\mathbf{k}}) := \int_{S^{d-1}} \sigma(\hat{\mathbf{k}} \cdot \hat{\mathbf{k}}') f(\hat{\mathbf{k}}') d\Omega(\hat{\mathbf{k}}')$ whenever $\langle \mathbf{e} \rangle = \mathbf{e}$.^{17,28} Indeed, for any linear transform R of the orthogo-

nal group $O(S^{d-1})$ leaving invariant the unit sphere S^{d-1} and its measure $d\Omega$, i.e., rotations and symmetries, one has

$$\begin{aligned}\mathcal{K}_0(\hat{\mathbf{k}} \cdot \mathbf{e}) &= \int_{S^{d-1}} \sigma(R\hat{\mathbf{k}} \cdot R\hat{\mathbf{k}}') \hat{\mathbf{k}}' \cdot \mathbf{e} d\Omega(\hat{\mathbf{k}}') \\ &= \int_{S^{d-1}} \sigma(R\hat{\mathbf{k}} \cdot \hat{\mathbf{k}}') \hat{\mathbf{k}}' \cdot R\mathbf{e} d\Omega(\hat{\mathbf{k}}'),\end{aligned}$$

using the change of variable $R\hat{\mathbf{k}}' \rightarrow \hat{\mathbf{k}}'$ and $R^{-1}=R^T$. Introducing $\mathbf{e}^\perp$ such that $\mathbf{e}=(\hat{\mathbf{k}} \cdot \mathbf{e})\hat{\mathbf{k}} + \mathbf{e}^\perp$ yields $\hat{\mathbf{k}}' \cdot R\mathbf{e} = (\hat{\mathbf{k}} \cdot \mathbf{e})R\hat{\mathbf{k}} \cdot \hat{\mathbf{k}}' + R\mathbf{e}^\perp \cdot \hat{\mathbf{k}}'$ with $\int_{S^{d-1}} \sigma(R\hat{\mathbf{k}} \cdot \hat{\mathbf{k}}') R\mathbf{e}^\perp \cdot \hat{\mathbf{k}}' d\Omega(\hat{\mathbf{k}}') = 0$ by symmetry (this can be seen by expressing this last integral in polar coordinates). Therefore,

$$\begin{aligned}\mathcal{K}_0(\hat{\mathbf{k}} \cdot \mathbf{e}) &= \int_{S^{d-1}} \sigma(R\hat{\mathbf{k}} \cdot \hat{\mathbf{k}}') (\hat{\mathbf{k}} \cdot \mathbf{e}) R\hat{\mathbf{k}} \cdot \hat{\mathbf{k}}' d\Omega(\hat{\mathbf{k}}') \\ &= \left(\int_{S^{d-1}} \sigma(\hat{\mathbf{k}} \cdot \hat{\mathbf{k}}') \hat{\mathbf{k}} \cdot \hat{\mathbf{k}}' d\Omega(\hat{\mathbf{k}}') \right) (\hat{\mathbf{k}} \cdot \mathbf{e}).\end{aligned}$$

Plugging Eq. (39) into Eq. (38) with $\mathbf{e} \equiv \nabla_{\mathbf{x}} \phi$ from the above analysis, Eq. (37), it is deduced that Λ^* satisfies

$$(\tilde{\Sigma} - \tilde{\Sigma}^*) \Lambda^* \mathbf{u} = \mathbf{C} \mathbf{u}, \quad (40)$$

with

$$\tilde{\Sigma}^*(\mathbf{x}, \omega) = \int_{S^{d-1}} \tilde{\sigma}(\mathbf{x}, \omega, \hat{\mathbf{k}} \cdot \hat{\mathbf{k}}') \hat{\mathbf{k}} \cdot \hat{\mathbf{k}}' d\Omega(\hat{\mathbf{k}}').$$

To the second order $O(\delta^2)$, we have

$$\begin{aligned}-\Omega_d \mathcal{Q}_\omega \mathbf{C}^{-d} \mathbf{a}^{(2)} &= \partial_t \mathbf{a}^{(0)} + \boldsymbol{\eta} \mathbf{a}^{(0)} + (\mathbf{C} \otimes \hat{\mathbf{k}}) \nabla_{\mathbf{x}} \otimes \mathbf{a}^{(1)} \\ &\quad - (\mathbf{C} \otimes \nabla_{\mathbf{x}}) (\mathbf{I}_d - \hat{\mathbf{k}} \otimes \hat{\mathbf{k}}) \nabla_{\hat{\mathbf{k}}} \otimes \mathbf{a}^{(1)},\end{aligned}$$

with $\boldsymbol{\eta}(\mathbf{x}) = \text{diag}(\eta_\alpha(\mathbf{x}))_{\alpha=1, \dots, M}$. The compatibility condition for $\mathbf{C}^{-d} \mathbf{a}^{(2)}$ to exist is that the right hand side in the above equation should belong to $N(\mathcal{Q}_\omega)^\perp$, that is, its averaged scalar product with any constant function $\phi \mathbf{d}$ should vanish. Therefore,

$$\begin{aligned}\mathbf{d} \cdot \langle \partial_t \mathbf{a}^{(0)} + (\mathbf{C} \otimes \hat{\mathbf{k}}) \nabla_{\mathbf{x}} \otimes \mathbf{a}^{(1)} - (\mathbf{C} \otimes \nabla_{\mathbf{x}}) (\mathbf{I}_d - \hat{\mathbf{k}} \otimes \hat{\mathbf{k}}) \\ \times \nabla_{\hat{\mathbf{k}}} \otimes \mathbf{a}^{(1)} + \boldsymbol{\eta} \mathbf{a}^{(0)} \rangle = 0,\end{aligned}$$

or, upon using Eq. (39) and reminding that neither $\mathbf{a}^{(0)}$ nor $\mathbf{a}^{(1)}$ depend on $\hat{\mathbf{k}}$:

$$\begin{aligned}\mathbf{d} \cdot \partial_t \mathbf{a}^{(0)} + \mathbf{d} \cdot \langle (\mathbf{C} \otimes \hat{\mathbf{k}}) \nabla_{\mathbf{x}} \otimes \mathbf{a}^{(11)} \rangle - \mathbf{d} \cdot \langle (\mathbf{C} \otimes \hat{\mathbf{k}}) \\ \times \nabla_{\mathbf{x}} \otimes (\Lambda^* \otimes \hat{\mathbf{k}}) \nabla_{\mathbf{x}} \otimes \mathbf{a}^{(0)} \rangle + \mathbf{d} \cdot \langle (\mathbf{C} \otimes \nabla_{\mathbf{x}}) \\ \times (\mathbf{I}_d - \hat{\mathbf{k}} \otimes \hat{\mathbf{k}}) \nabla_{\hat{\mathbf{k}}} \otimes (\Lambda^* \otimes \hat{\mathbf{k}}) \nabla_{\mathbf{x}} \otimes \mathbf{a}^{(0)} \rangle \\ + \mathbf{d} \cdot \langle \boldsymbol{\eta} \mathbf{a}^{(0)} \rangle = 0.\end{aligned}$$

Here it is assumed that $\langle \boldsymbol{\eta} \rangle \neq \boldsymbol{\eta}$ for generality purposes, as $\boldsymbol{\eta}$ may also depend on $\mathbf{k}$. But by definition of the product of a third order tensor and a second order tensor $(\mathbf{A} \otimes \mathbf{e}) \mathbf{f} \otimes \mathbf{g} = (\mathbf{e} \cdot \mathbf{f}) \mathbf{A} \mathbf{g}$, one first has

$$\mathbf{d} \cdot \langle (\mathbf{C} \otimes \hat{\mathbf{k}}) \nabla_{\mathbf{x}} \otimes \mathbf{a}^{(11)} \rangle = \mathbf{d} \cdot \mathbf{C} \langle \hat{\mathbf{k}} \cdot \nabla_{\mathbf{x}} \rangle \mathbf{a}^{(11)} = 0$$

because $\langle \hat{\mathbf{k}} \cdot \nabla_{\mathbf{x}} \rangle = 0$. Second,

$$\begin{aligned}\langle (\mathbf{C} \otimes \hat{\mathbf{k}}) \nabla_{\mathbf{x}} \otimes (\Lambda^* \otimes \hat{\mathbf{k}}) \nabla_{\mathbf{x}} \otimes \mathbf{a}^{(0)} \rangle \\ = \mathbf{C} \langle (\hat{\mathbf{k}} \cdot \nabla_{\mathbf{x}}) \Lambda^* (\hat{\mathbf{k}} \cdot \nabla_{\mathbf{x}}) \rangle \mathbf{a}^{(0)} = \frac{1}{d} \mathbf{C} \nabla_{\mathbf{x}} \cdot \Lambda^* (\mathbf{a}^{(0)} \otimes \nabla_{\mathbf{x}})\end{aligned}$$

because $\langle (\hat{\mathbf{k}} \cdot \nabla_{\mathbf{x}}) f(\hat{\mathbf{k}} \cdot \nabla_{\mathbf{x}}) \rangle = (1/d) \nabla_{\mathbf{x}} \cdot f \nabla_{\mathbf{x}}$. Finally,

$$\begin{aligned}\langle (\mathbf{C} \otimes \nabla_{\mathbf{x}}) (\mathbf{I}_d - \hat{\mathbf{k}} \otimes \hat{\mathbf{k}}) \nabla_{\hat{\mathbf{k}}} \otimes (\Lambda^* \otimes \hat{\mathbf{k}}) \nabla_{\mathbf{x}} \otimes \mathbf{a}^{(0)} \rangle \\ = (\mathbf{C} \otimes \nabla_{\mathbf{x}}) \langle (\mathbf{I}_d - \hat{\mathbf{k}} \otimes \hat{\mathbf{k}}) \nabla_{\hat{\mathbf{k}}} \otimes \Lambda^* (\hat{\mathbf{k}} \cdot \nabla_{\mathbf{x}}) \rangle \mathbf{a}^{(0)} \\ = (\mathbf{C} \otimes \nabla_{\mathbf{x}}) \langle \mathbf{I}_d - \hat{\mathbf{k}} \otimes \hat{\mathbf{k}} \rangle \Lambda^* (\mathbf{a}^{(0)} \otimes \nabla_{\mathbf{x}}) \\ = \frac{d-1}{d} (\mathbf{C} \otimes \nabla_{\mathbf{x}}) \cdot \Lambda^* (\mathbf{a}^{(0)} \otimes \nabla_{\mathbf{x}}),\end{aligned}$$

since $\nabla_{\hat{\mathbf{k}}} \otimes \hat{\mathbf{k}} = \mathbf{I}_d$ and $\langle \mathbf{I}_d - \hat{\mathbf{k}} \otimes \hat{\mathbf{k}} \rangle = (1 - (1/d)) \mathbf{I}_d$. Therefore $\mathbf{a}^{(0)}$ satisfies the diffusion equation:

$$\begin{aligned}\mathbf{d} \cdot (\partial_t \mathbf{a}^{(0)} - \nabla_{\mathbf{x}} \cdot \mathbf{D} (\mathbf{a}^{(0)} \otimes \nabla_{\mathbf{x}}) + (\mathbf{C} \otimes \nabla_{\mathbf{x}}) \cdot \Lambda^* \\ \times (\mathbf{a}^{(0)} \otimes \nabla_{\mathbf{x}}) + \langle \boldsymbol{\eta} \rangle \mathbf{a}^{(0)}) = 0,\end{aligned} \quad (41)$$

where the diffusion matrix $\mathbf{D}$ is

$$\mathbf{D}(\mathbf{x}, \omega) = \frac{1}{d} \mathbf{C}(\mathbf{x}) \Lambda^*(\mathbf{x}, \omega).$$

This result extends the conclusions of Refs. 32–34 to the case of heterogeneous background media with multimode diffusion. The main advantage of the diffusion equation is that its solution is independent of the direction $\hat{\mathbf{k}}$. Its numerical solution is therefore much faster to construct than that of the full multigroup transport equations (31). Alternatively, Eq. (41) may be written in a conservative form for the constant function ϕ of Eq. (37) as:³⁵

$$\frac{1}{\tilde{c}^d(\mathbf{x})} \partial_t \phi - \nabla_{\mathbf{x}} \cdot \tilde{D}(\mathbf{x}, \omega) \nabla_{\mathbf{x}} \phi + \tilde{\eta}(\mathbf{x}, \omega) \phi = 0, \quad (42)$$

where $\tilde{c} = \|\mathbf{C}\|_{-d}$, $\tilde{D} = \|\mathbf{C}^{-d} \mathbf{D}\|_1$ and $\tilde{\eta} = \|\mathbf{C}^{-d} \langle \boldsymbol{\eta} \rangle\|_1$:

$$\begin{aligned}\tilde{c}(\mathbf{x}) &= \left(\sum_{\alpha=1}^M c_\alpha^{-d}(\mathbf{x}) \right)^{-1/d}, \quad \tilde{D}(\mathbf{x}, \omega) = \frac{1}{d} \sum_{\alpha=1}^M \frac{\ell_\alpha^*(\mathbf{x}, \omega)}{c_\alpha^{d-1}(\mathbf{x})}, \\ \tilde{\eta}(\mathbf{x}, \omega) &= \sum_{\alpha=1}^M \frac{\langle \eta_\alpha \rangle(\mathbf{x}, \omega)}{c_\alpha^d(\mathbf{x})}.\end{aligned}$$

The so-called transport mean free paths $(\ell_1^*, \ell_2^*, \dots, \ell_M^*)^T = \Lambda^* \mathbf{u}$ are computed by Eq. (40). They characterize the distance waves must propagate until their direction $\hat{\mathbf{k}}$ is randomized. Generally $\ell_\alpha^* > \ell_\alpha$ for a tendency of forward scattering such that the matrix $\tilde{\Sigma}^*$ is positive definite. If scattering is uniform, then $\tilde{\Sigma}^* = \mathbf{0}$ and $\ell_\alpha^* = \ell_\alpha$ for $\alpha = 1, 2, \dots, M$.

C. Energy density equipartition

Using the equipartition rule (37), the space-time energy density of Eq. (32) in the limit $\delta \rightarrow 0$ is

$$\mathcal{E}^{(0)}(\mathbf{x}, t) = \frac{1}{\tilde{c}_\alpha^d(\mathbf{x})} \int_0^{+\infty} e^{(0)}(\mathbf{x}, \omega, t) \omega^{d-1} d\omega, \quad (43)$$

$$e^{(0)}(\mathbf{x}, \omega, t) = \Omega_d \phi(\mathbf{x}, \omega, t).$$

Introducing the energy density $\mathcal{E}_\alpha^{(0)}$ of a particular mode α among the m existing ones, such that $\mathcal{E}^{(0)} = \sum_{\alpha=1}^m R_\alpha \mathcal{E}_\alpha^{(0)}$, where the α th mode has an order of multiplicity R_α , one typically has for $t > \ell_\alpha^*/c_\alpha$:

$$\frac{\mathcal{E}_\alpha^{(0)}(\mathbf{x}, t)}{\mathcal{E}_\beta^{(0)}(\mathbf{x}, t)} = \frac{R_\alpha}{R_\beta} \left(\frac{c_\beta(\mathbf{x})}{c_\alpha(\mathbf{x})} \right)^d, \quad \forall \alpha, \beta = 1, 2, \dots, m, \quad (44)$$

$$\forall \mathbf{x} \in \mathcal{O},$$

whatever the underlying random heterogeneity of the medium or the loads may be.

D. Fick's law

The “transmission relationship” or “energetic behavior law” which links the power flow with the gradient of the energy density, as invoked in the VCA literature, is derived from the previous results as follows. Taking the scalar product of Eq. (39) with $\mathbf{C}^{1-d} \mathbf{u}$, then premultiplying it with $\hat{\mathbf{k}}$ and integrating over S^{d-1} [see Eq. (33)], yields the space-time power flow density vector as

$$\Pi^\delta(\mathbf{x}, t) = \sum_{\alpha=1}^M \frac{1}{c_\alpha^d(\mathbf{x})} \int_0^{+\infty} \pi_\alpha^\delta(\mathbf{x}, \omega, t) \omega^{d-1} d\omega, \quad (45)$$

$$\pi_\alpha^\delta(\mathbf{x}, \omega, t) = -\delta D_\alpha(\mathbf{x}, \omega) \nabla_\mathbf{x} e^{(0)} + o(\delta),$$

where $D_\alpha(\mathbf{x}, \omega) = (1/d) c_\alpha(\mathbf{x}) \ell_\alpha^*(\mathbf{x}, \omega)$ is a partial diffusion coefficient. Thus $\Pi^\delta(\mathbf{x}, t) = \mathbf{0}$ in the limit $\delta \rightarrow 0$, as expected for a diffusive regime. Recalling Eq. (36) and (39), one has the expansion:

$$a_\alpha^\delta(\mathbf{x}, \omega, \hat{\mathbf{k}}, t) = \frac{1}{\Omega_d} \left[e^{(0)}(\mathbf{x}, \omega, t) + \frac{d}{c_\alpha^2(\mathbf{x})} \mathbf{c}_{g,\alpha}(\mathbf{x}, \mathbf{k}) \cdot \nabla_\mathbf{x} e^{(0)}(\mathbf{x}, \omega, t) \right] + O(\delta^2), \quad (46)$$

which is, ignoring the $O(\delta^2)$ terms, the usual P_1 -approximation invoked in neutron transport theory^{16,28} in order to derive diffusion approximations from the lowest-order moment equations induced by Eq. (31).

E. Examples

The scalar case $m=M=1$ has been considered in Ref. 35. The diffusion equation (41) reduces to

$$\partial_t a^{(0)} - \nabla_\mathbf{x} \cdot D(\mathbf{x}, \omega) \nabla_\mathbf{x} a^{(0)} + \ell^*(\mathbf{x}, \omega) \nabla_\mathbf{x} c \cdot \nabla_\mathbf{x} a^{(0)} + \langle \eta \rangle(\mathbf{x}, \omega) a^{(0)} = 0, \quad (47)$$

where the diffusion coefficient D and the transport mean free path ℓ^* are

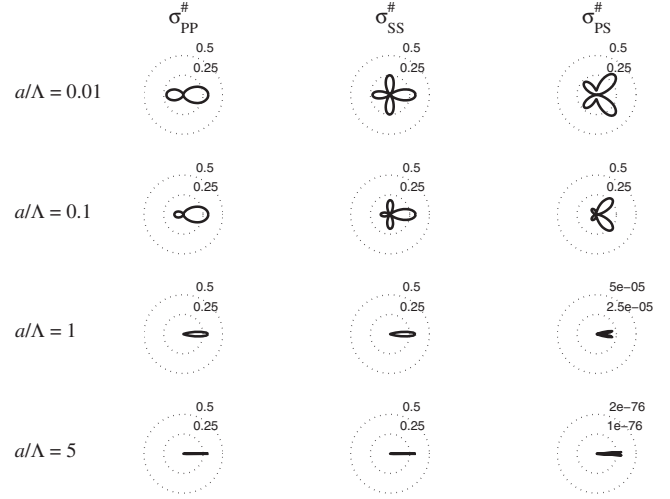


FIG. 2. Normalized differential scattering cross-sections $\sigma_{\alpha\beta}^\# = (\Lambda/ac_\omega)^2 \sigma_{\alpha\beta}$ for a thick shell as functions of $\Theta = \arccos(\hat{\mathbf{k}} \cdot \hat{\mathbf{k}}')$ for various ratio a/Λ and Gaussian correlation model; $\nu=0.2$, $r_{11}=r_{12}=r_{13}=0.1$, $r_{22}=r_{33}=1.0$, and $r_{23}=0.9$.

$$D(\mathbf{x}, \omega) = \frac{1}{d} c(\mathbf{x}) \ell^*(\mathbf{x}, \omega),$$

$$\ell^*(\mathbf{x}, \omega) = c(\mathbf{x}) \left(\int_{S^{d-1}} \tilde{\sigma}(\mathbf{x}, \omega, \hat{\mathbf{k}} \cdot \hat{\mathbf{k}}') \times (1 - \hat{\mathbf{k}} \cdot \hat{\mathbf{k}}') d\Omega(\hat{\mathbf{k}}') \right)^{-1},$$

respectively. In a conservative form, the diffusion equation is

$$\frac{1}{c^d(\mathbf{x})} \partial_t a^{(0)} - \nabla_\mathbf{x} \cdot \left(\frac{D(\mathbf{x}, \omega)}{c^d(\mathbf{x})} \nabla_\mathbf{x} a^{(0)} \right) + \frac{\langle \eta \rangle(\mathbf{x}, \omega)}{c^d(\mathbf{x})} a^{(0)} = 0.$$

Note that it is also applicable to the phase-space energy density $w^{(0)}$ or to the space-time energy density $\mathcal{E}^{(0)}$ (in the limit $\delta \rightarrow 0$) of Eq. (43):³⁰

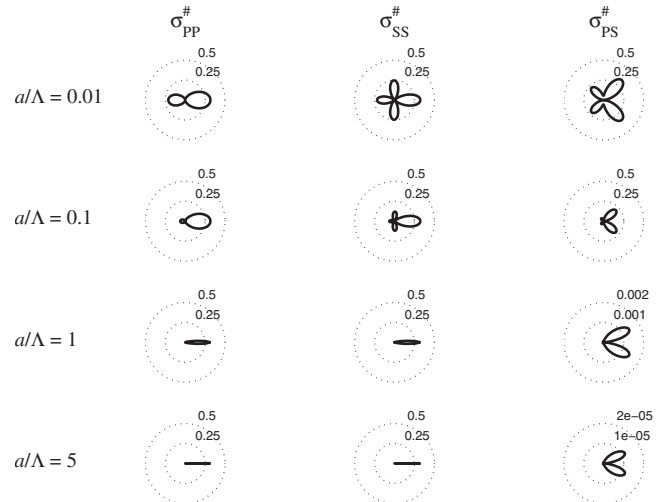


FIG. 3. Normalized differential scattering cross sections $\sigma_{\alpha\beta}^\# = (\Lambda/ac_\omega)^2 \sigma_{\alpha\beta}$ for a thick shell as functions of $\Theta = \arccos(\hat{\mathbf{k}} \cdot \hat{\mathbf{k}}')$ for various ratio a/Λ and exponential (Markov) correlation model; same parameters as for Fig. 2.

$$\mathcal{E}^{(0)}(\mathbf{x}, t) = \frac{\Omega_d}{c^d(\mathbf{x})} \int_0^{+\infty} a^{(0)}(\mathbf{x}, \omega, t) \omega^{d-1} d\omega.$$

This example covers the diffusive regime for shear and bending energy densities in Timoshenko beams with $d=1$ (Sec. II B) and for the transverse energy density in thick shells with $d=2$ (Sec. II A).

For the in-plane and bending energy densities of a thick

$$\begin{pmatrix} \ell_P^* \\ \ell_S^* \end{pmatrix} = \begin{bmatrix} \tilde{\Sigma}_{PP}\left(\mathbf{x}, \frac{\omega}{c_P}\right) + \tilde{\Sigma}_{PS}\left(\mathbf{x}, \frac{\omega}{c_P}\right) - \tilde{\Sigma}_{PP}^*\left(\mathbf{x}, \frac{\omega}{c_P}\right) & -\tilde{\Sigma}_{PS}^*\left(\mathbf{x}, \frac{\omega}{c_P}\right) \\ -\tilde{\Sigma}_{SP}^*\left(\mathbf{x}, \frac{\omega}{c_S}\right) & \tilde{\Sigma}_{SS}\left(\mathbf{x}, \frac{\omega}{c_S}\right) + \tilde{\Sigma}_{SP}\left(\mathbf{x}, \frac{\omega}{c_S}\right) - \tilde{\Sigma}_{SS}^*\left(\mathbf{x}, \frac{\omega}{c_S}\right) \end{bmatrix}^{-1} \begin{pmatrix} c_P(\mathbf{x}) \\ c_S(\mathbf{x}) \end{pmatrix},$$

with

$$\begin{aligned} \tilde{\Sigma}_{\alpha\beta}^*\left(\mathbf{x}, \frac{\omega}{c_\alpha}\right) &= \left(\frac{c_\alpha}{c_\beta}\right)^2 \tilde{\Sigma}_{\beta\alpha}^*\left(\mathbf{x}, \frac{\omega}{c_\beta}\right) \\ &= \int_{S^1} \tilde{\sigma}_{\alpha\beta}\left(\mathbf{x}, \frac{\omega}{c_\alpha}, \frac{\omega}{c_\beta}, \hat{\mathbf{k}} \cdot \hat{\mathbf{k}}'\right) \hat{\mathbf{k}} \cdot \hat{\mathbf{k}}' d\Omega(\hat{\mathbf{k}}'), \\ \alpha, \beta &= P, S. \end{aligned}$$

If the background medium is homogeneous, such that the wave speeds c_α are independent of $\mathbf{x}$ for $\alpha=1, 2, \dots, M$, the diffusion equation reduces to

$$\mathbf{d} \cdot (\partial_t \mathbf{a}^{(0)} - \nabla_{\mathbf{x}} \cdot \mathbf{D}(\mathbf{a}^{(0)} \otimes \nabla_{\mathbf{x}}) + \langle \boldsymbol{\eta} \rangle \mathbf{a}^{(0)}) = 0.$$

Its solution also satisfies the equipartition rule (37), that is, $\mathbf{a}^{(0)} = \phi \mathbf{u}$ where ϕ is a constant function of $\hat{\mathbf{k}}$ ($\langle \phi \rangle = \phi$) which solves

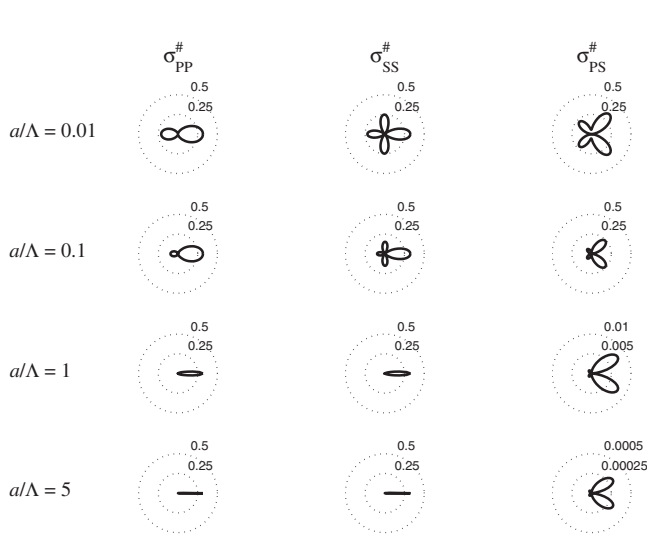


FIG. 4. Normalized differential scattering cross-sections $\sigma_{\alpha\beta}^{\#} = (\Lambda/ac_\alpha)^2 \sigma_{\alpha\beta}$ for a thick shell as functions of $\Theta = \arccos(\hat{\mathbf{k}} \cdot \hat{\mathbf{k}}')$ for various ratio a/Λ and Von Karman correlation model; same parameters as for Fig. 2.

shell, $m=2$ and $M=4$ since each mode $m=P$ or $m=S$ has multiplicity 2 (see Sec. II A). The diffusion constant $\tilde{D}_{PS}$ in Eq. (42) for this group of coupled energy modes is

$$\tilde{D}_{PS}(\mathbf{x}, \omega) = \frac{1}{2} \left(\frac{\ell_P^*(\mathbf{x}, \omega)}{c_P(\mathbf{x})} + \frac{\ell_S^*(\mathbf{x}, \omega)}{c_S(\mathbf{x})} \right),$$

where the transport mean free paths ℓ_P^* and ℓ_S^* are given by Eq. (40):

$$\partial_t \phi - \tilde{D} \Delta_{\mathbf{x}} \phi + \tilde{\eta} \phi = 0, \quad (48)$$

where

$$\tilde{D}(\omega) = \frac{\sum_{\alpha=1}^M c_\alpha^{1-d} \ell_\alpha^*(\omega)}{d \sum_{\alpha=1}^M c_\alpha^{-d}}, \quad \tilde{\eta}(\omega) = \frac{\sum_{\alpha=1}^M c_\alpha^{-d} \langle \eta_\alpha \rangle(\omega)}{\sum_{\alpha=1}^M c_\alpha^{-d}},$$

as in Refs. 32 and 33 for slender structures or Ref. 34 for three-dimensional media. For a thick shell the diffusion constant of the PS group is

$$\tilde{D}_{PS}(\omega) = \frac{c_P c_S}{2(c_P^2 + c_S^2)} (c_S \ell_P^*(\omega) + c_P \ell_S^*(\omega)),$$

and Fick's law writes

$$\boldsymbol{\pi}_{PS}^\delta(\mathbf{x}, \omega, t) = -\delta \tilde{D}_{PS}(\omega) \nabla_{\mathbf{x}} e_{PS}^{(0)}(\mathbf{x}, \omega, t) + o(\delta). \quad (49)$$

Figures 2–5 display the normalized differential scattering

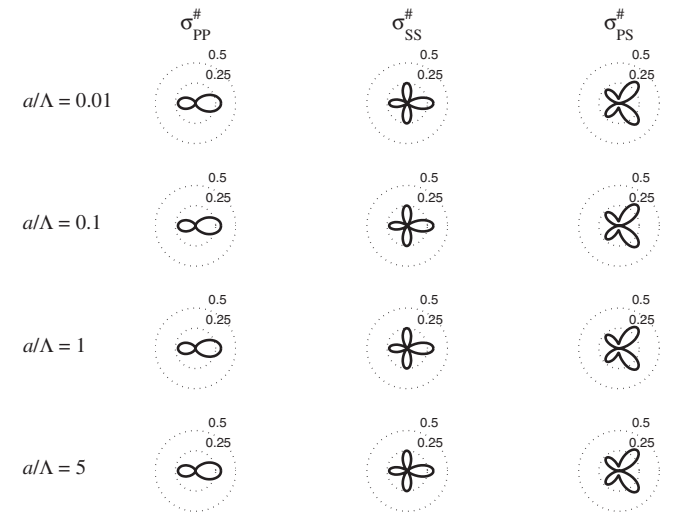


FIG. 5. Normalized differential scattering cross-sections $\sigma_{\alpha\beta}^{\#} = (\Lambda/ac_\alpha)^2 \sigma_{\alpha\beta}$ for a thick shell as functions of $\Theta = \arccos(\hat{\mathbf{k}} \cdot \hat{\mathbf{k}}')$ for various ratio a/Λ and delta (Rayleigh) correlation model; same parameters as for Fig. 2.

TABLE I. Correlation models; K_0 is the zeroth order modified Bessel function.

Correlation function	$N(\mathbf{x})$	$\hat{N}(\mathbf{k})$
Gaussian	$\exp\left(-\frac{ \mathbf{x} ^2}{2a^2}\right)$	$\frac{a^2}{2\pi} \exp\left(-\frac{1}{2}a^2 \mathbf{k} ^2\right)$
Markov	$\exp\left(-\frac{ \mathbf{x} }{a}\right)$	$\frac{a^2}{2\pi} (1+a^2 \mathbf{k} ^2)^{-3/2}$
von Karman	$K_0\left(\frac{ \mathbf{x} }{a}\right)$	$\frac{a^2}{2\pi} (1+a^2 \mathbf{k} ^2)^{-1}$
Rayleigh	$\delta\left(\frac{ \mathbf{x} }{a}\right)$	$\frac{a^2}{2\pi^2}$

cross sections $\sigma_{\alpha\beta}^{\#} = (\Lambda/ac_{\alpha})^2 \sigma_{\alpha\beta}$ of a shell as functions of $\Theta = \arccos(\hat{\mathbf{k}} \cdot \hat{\mathbf{k}}')$ for four different models of correlation functions parametrized by a characteristic length a of random heterogeneities, and for different values of the ratio a/Λ with $\Lambda = 2\pi c_P/\omega$. The correlation functions are defined by $R_{jk}(\mathbf{x}-\mathbf{y}) = \int_{\mathbb{R}^2} e^{-i\mathbf{k} \cdot (\mathbf{x}-\mathbf{y})} S_{jk}(\mathbf{k}) d\mathbf{k}$ and the four models are gathered in Table I below. They have the form $R_{jk}(\mathbf{x}) = r_{jk} N(\mathbf{x})$, where r_{jk} is a correlation coefficient and $N(\mathbf{x})$ is the normalized correlation function, such that $S_{jk}(\mathbf{k}) = r_{jk} \hat{N}(\mathbf{k})$ and $\hat{N}(\mathbf{k})$ stands for the two-dimensional Fourier transform of $N(\mathbf{x})$. The parameters chosen for all computations are $\nu = 1 - 2(c_S^2/c_P^2) = 0.2$ (mean Poisson's ratio), $r_{11} = 0.1$ and $r_{22} = r_{33} = 1$ (small fluctuations of the density versus Lamé's moduli), and $r_{12} = r_{13} = 0.1$ and $r_{23} = 0.9$ (weak correlation of the density with Lamé's moduli). It is observed that for the first three models the scattering cross sections get strongly forward peaked as the ratio a/Λ increases, and scattering is mostly elastic. Indeed, the inelastic scattering process corresponding to mode conversions $P \leftrightarrow S$ as quantified by σ_{PS} or σ_{SP} becomes negligible. A noteworthy exception is the delta (Rayleigh) correlation model for which neither elastic or inelastic scattering dominates, and backscattering as well as S -transverse scattering subsist even at the higher ratios a/Λ ; delta-correlated scatterers may therefore be more amenable to localize than others. As a global trend, the amounts of inelastic scattering and backscattering increase with the roughness of the correlation function. Next, the normalized scattering and transport mean free paths ℓ_{α}/Λ and ℓ_{α}^*/Λ are plotted in Figs. 6 and 7 below for modes $\alpha = P$ and $\alpha = S$, respectively, as functions of a/Λ . Figure 8 displays the normalized diffusion constant $\tilde{D}_{PS}/c_P\Lambda$ for the same models of correlation functions and parameters. Note that the normalization chosen here for the figures is different from Refs. 33 and 34. These results agree remarkably well with the ones reported in Ref. 45 for diffuse light propagation in porous glasses in the range $0 < a/\Lambda < 1$, both qualitatively and quantitatively for the exponential (Markov) model. It may also be observed that they are only slightly affected by variations in the correlation parameters r_{jk} (plots not displayed here). The scattering mean free paths are almost equal for the first three models; however, the transport mean free paths differ significantly from one model to another. One may observe from these results that smooth perturbations (Gaussian model) tend to delay the onset of the diffusion regime compared to sharper perturbations (other models). Also, the

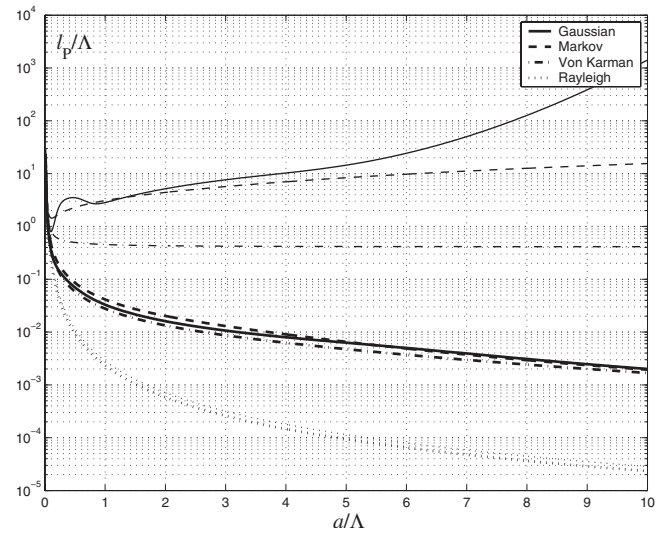


FIG. 6. Normalized scattering (thick lines) and transport (thin lines) P -mode mean free paths ℓ_P/Λ for a thick shell as functions of a/Λ for various models of correlation; same parameters as for Fig. 2.

sharper the perturbations are, the smaller the diffusion constant is, indicating a possible prominence of weak localization at the higher frequencies. For example, the diffusion constant with a delta correlation function apparently scales inversely with the correlation length a , in accordance with Thouless' assertion.

V. SUMMARY

The elastic energy and power flow densities $\mathcal{E}^{\epsilon}(\mathbf{x}, t)$ and $\Pi^{\epsilon}(\mathbf{x}, t)$, respectively, e.g., Eqs. (5) and (6) for shells, satisfy the continuity equation:

$$\partial_t \mathcal{E}^{\epsilon} + \nabla_{\mathbf{x}} \cdot \Pi^{\epsilon} = 0,$$

for any fixed $\epsilon > 0$, up to a possible dissipative term. However, the passage to the limit $\epsilon \rightarrow 0$ is difficult because it requires the estimation of the limit of the *square* of highly oscillating functions, together with the passage to the limit in

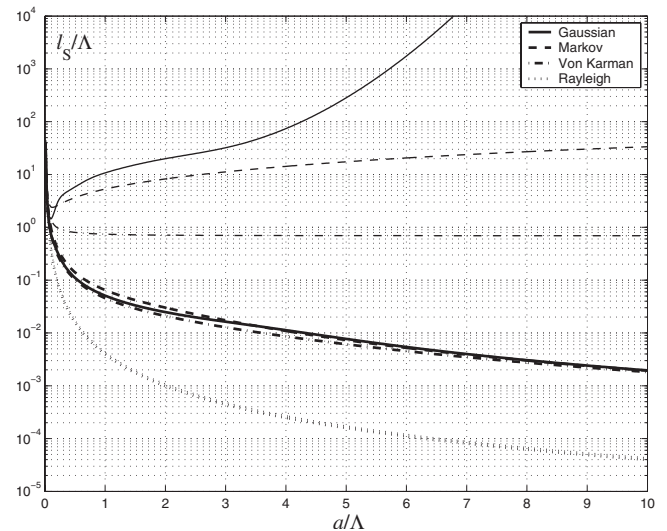


FIG. 7. Normalized scattering (thick lines) and transport (thin lines) S -mode mean free paths ℓ_S/Λ for a thick shell as functions of a/Λ for various models of correlation; same parameters as for Fig. 2.

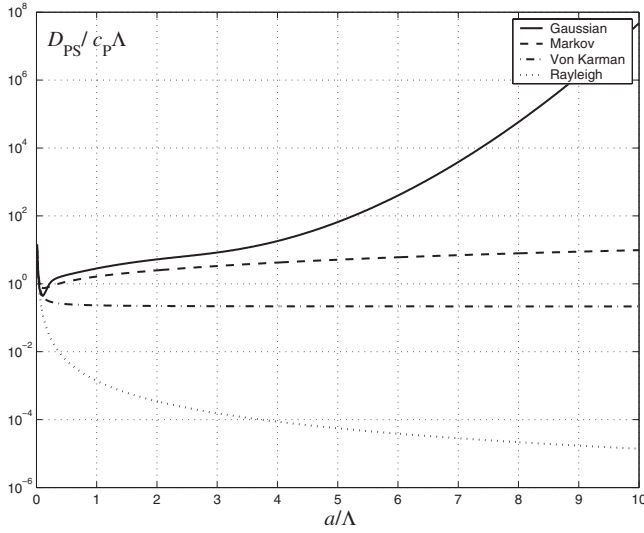


FIG. 8. Normalized diffusion constant $\tilde{D}_{PS}/c_p\Lambda$ for a thick shell as function of a/Λ for various models of correlation; same parameters as for Fig. 2.

the above partial differential equation. Microlocal analysis and kinetic models describe how this should be done, as expounded in Sec. II for slender structures. In turn, the multigroup transport equations (35) in the near critical diffusion scaling correspond to a continuity equation:

$$\partial_t e^\delta + \frac{1}{\delta} \nabla_{\mathbf{x}} \cdot \boldsymbol{\pi}^\delta + P^\delta = 0,$$

for any fixed $\delta > 0$, where

$$e^\delta(\mathbf{x}, \omega, t) = \sum_{\alpha=1}^M c_\alpha^{-d}(\mathbf{x}) \int_{S^{d-1}} a_\alpha^\delta(\mathbf{x}, \omega, \hat{\mathbf{k}}, t) d\Omega(\hat{\mathbf{k}}),$$

$$\boldsymbol{\pi}^\delta(\mathbf{x}, \omega, t) = \sum_{\alpha=1}^M c_\alpha^{-d}(\mathbf{x}) \int_{S^{d-1}} \mathbf{c}_{g,\alpha}(\mathbf{x}, \mathbf{k}) a_\alpha^\delta(\mathbf{x}, \omega, \hat{\mathbf{k}}, t) d\Omega(\hat{\mathbf{k}}),$$

and

$$P^\delta(\mathbf{x}, \omega, t) = \sum_{\alpha=1}^M c_\alpha^{-d}(\mathbf{x}) \int_{S^{d-1}} \eta_\alpha(\mathbf{x}, \mathbf{k}) a_\alpha^\delta(\mathbf{x}, \omega, \hat{\mathbf{k}}, t) d\Omega(\hat{\mathbf{k}}),$$

for the dissipated power density. Its limit $\delta \rightarrow 0$ is the diffusion equation (42), which may be recovered formally from the continuity equation above and Fick's law (45). This is what is done in the VCA literature,¹⁹ ignoring, however, the essential phase dependence of the energy and power flow densities before diffusion sets up, and all possible interactions between different modes. As already noted in Sec. IV D, a crude derivation of the diffusion equation may also be carried on resorting to the so-called P_1 -approximation, which corresponds to the lowest-order spherical harmonics expansion of the specific intensities. This approach is used in Ref. 38 to formally derive "generalized" diffusion equations (of the form obtained in Ref. 29) using several additional assumptions, including a Fick-like law. The present analysis allows to estimate the error induced by this assumption and a P_1 -approximation, i.e., Eqs. (45) and (46).

The universal equipartition rule of Eq. (44) can be used as an indicator of the relevance of the diffusive approxima-

tion, the latter being acceptable as soon as this relation is satisfied by the energy densities up to an acceptable accuracy level. It should be noted incidentally that this relation is fully compatible with the assumption of *modal* energy equipartition in SEA since modal densities of bounded media are proportional to the inverse wave speed to the power of dimension d to the leading order versus frequency. Another indicator of a diffusive behavior is the fact that power flows vanish in the limit of small scattering mean free paths, see Eq. (45), as a consequence of the strong mixing of wave directions induced by multiple scattering. Having in mind that energy fluxes are roughly proportional to the gradient of the wave phases, high-frequency vibrations should exhibit smooth, indeed constant, phase variations for different receivers. This trend has been clearly observed on the complex structure studied in Ref. 27, among others. For this system, one must invoke the many reflections of the waves on stiffeners, bulkheads, and other structural components (the scatterers) to justify a diffuse vibrational energy field after some time. Then it should always be possible to estimate a diffusion constant for such a structure, as done for complex enclosures in room acoustics. That possibility shall be investigated in future works.

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